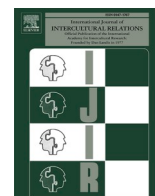




Contents lists available at ScienceDirect

International Journal of Intercultural Relations

journal homepage: www.elsevier.com/locate/ijintrel

Review

International student graduates' workforce integration: A systematic review

Yu Han^{a,*}, Daniel Gulanowski^b, Greg J. Sears^c^a Faculty of Business Administration University of Regina, 3737 Wascana Pkwy, Regina, SK S4S 0A2, Canada^b Spott School of Business Carleton University, 1125 Colonel By Drive Ottawa, Ontario K1S 5B6, Canada^c Spott School of Business Carleton University, 1125 Colonel By Drive Ottawa, Ontario K1S 5B6, Canada

ARTICLE INFO

Keywords:

International student graduate
Workforce integration
Systematic review

ABSTRACT

Although international students play a central role in addressing labour shortages and strengthening the economies of receiving countries, research examining the workforce integration of international graduates has been relatively sparse. Recently, however, this topic has garnered increasing empirical attention spanning across disciplines and levels of analysis. We conducted a systematic review of 74 academic articles examining the workforce integration of international graduates to take stock of the literature in this domain, summarize emergent themes and key findings, and highlight potential pathways for future research. Based on our review, we identify various antecedents of workforce integration at various levels of analysis (micro-, meso-, macro-), ranging from specific characteristics of the international graduate (e.g., host country cultural knowledge, language fluency, motivation, family status) to organizational attributes and practices (e.g., recruitment and selection, post-secondary institution support) to broader societal factors (e.g., immigration and labour policy/legislation, economic conditions), and delineate several promising avenues for future research.

Introduction

Facing heightened worldwide competition for talent, retaining and integrating international student graduates is critical for organizations and host countries (Mok & Chan, 2020; Shumilova & Cai, 2016). International students represent a significant part of the global talent pool and play a pivotal role in addressing growing labour shortages, increasing productivity and innovation, and boosting the economies of receiving countries (Hawthorne, 2018; Wilkinson, 2008). As a result, more countries are recognizing that international graduates are ideal immigrant candidates and are actively promoting their settlement and integration (Esses, Sutter, Ortiz, Luo, Cui, & Deacon, 2018; Suter & Jandl, 2008).

The number of international students is increasing significantly, with over 5.6 million international students worldwide in 2019, up from 2 million in 2000 (OECD, 2020; UNESCO, 2019). In 2019, the top three largest higher education host destinations were the United States, the United Kingdom, and Canada; together, they hosted approximately 39% of all international students (IIE, 2020). The economic impact of international students is significant for host countries. For example, the 1.1 million international students enrolled in the US contributed over \$45 billion to the US economy and they created or supported nearly 460,000 jobs (Educationdata, 2020). In Canada, 638,960 international students accounted for all of the growth in postsecondary enrolment in the 2018/2019 academic year

* Corresponding author.

E-mail addresses: yu.han@uregina.ca (Y. Han), daniel.gulanowski@carleton.ca (D. Gulanowski), greg.sears@carleton.ca (G.J. Sears).

(Statistics Canada, 2020). In 2021, international student intake in Canada declined by about 17% due to the global pandemic (Statista, 2021a); however, international students still represent a substantial part of total postsecondary enrolment (530,540 out of 2.15 million; Statista, 2021b). Despite increasing numbers of international students studying abroad, coupled with a high demand for highly educated and skilled talent in host countries, many international graduates face significant challenges finding suitable employment and integrating into host societies (Esses et al., 2018; Moskal, 2017).

The research literature on international students' workforce integration is fragmented and scattered across multiple disciplines, including management, education, migration, sociology, and psychology, among others. While international students have been widely studied in the higher education literature, the focus has mainly been on their academic, sociocultural, and psychological adjustment while completing their education (Gümüç et al., 2020). For example, studies have shown that international students face challenges in their adjustment that can negatively impact their educational experience and their socialization in the host country (Franco, Hsiao, Gnillka, & Ashby, 2019; Lashari et al., 2018; Ra & Trusty, 2017). Recent research has also shown that, due to travel restrictions and heightened uncertainty and anxiety triggered by the COVID-19 pandemic, international students have reported lower levels of well-being, and challenges with respect to their mobility and finding and securing employment (e.g., Firang, 2020; Hari et al., 2021; Jenei, Cassidy-Matthews, Virk, Lulie, & Closson, 2020). Despite this evidence that international students face various challenges when studying abroad, there is a paucity of research that has examined the postgraduation employment integration of international students in receiving countries.

Recent reviews have examined the employment and workforce integration of other migrant groups, including skilled migrants (e.g., Hajro, Stahl, Clegg, & Lazarova, 2019; Tharenou & Kulik, 2020) and refugees (Lee, Szkudlarek, Nguyen, & Nardon, 2020); however, these syntheses have not specifically examined the workforce integration of international graduates. Considering the critical importance of international students for organizations and host countries, coupled with data suggesting that the number of international graduates seeking employment in several host countries has risen in recent years (Choi, Crossman, & Hou, 2021; Morris-Lange, 2019; Ruiz & Budiman, 2018), a systematic review of the literature examining the workforce integration of international graduates is long overdue. Thus, this review aims to identify and synthesize research examining international student post-graduation work experiences and outcomes to better understand the state of current research in this area, and identify critical gaps and directions for future research. Based on our review, we identify key antecedents of international graduate workforce integration¹ relating to various levels of analysis (micro, meso, macro) and discuss several avenues for future research. Given the multidisciplinary nature of the subject matter, we found that different research methodologies have been applied in studying the workforce integration of international graduates (e.g., qualitative and quantitative); however, in terms of content, some topics have garnered significantly more attention than others. For example, several studies have examined migrant career outcomes using objective measures of success (e.g., earnings), and have compared international graduates to other groups. However, few studies have tracked this information over time or systematically assessed more subjective elements of international graduates' work integration (e.g., perceptions of workplace inclusion, career satisfaction, organizational support, quality of mentorship/supervision). Likewise, although there is evidence that international graduates may face various barriers to their integration (e.g., discrimination) when applying for jobs and during their post-hire work experience (Arthur & Flynn, 2013; Coffey et al., 2018; Peng, 2020), very little research has explored what specific practices organizations and educational institutions can implement to reduce these impediments and enhance the workplace integration of international graduates.

In the next section, we discuss the methodology used for this review. Based on our review of the literature, we then provide an overview of the key themes that emerged, and in particular, the primary individual (micro), organizational (meso), and societal-industry (macro) factors that influence international graduates' work experience and integration success. Finally, we outline suggestions for future research.

Literature search

This review focuses specifically on research articles examining international students' post-study work experience in a receiving country. To search for relevant articles, we used electronic databases in business management, psychology and sociology, including Business Source Complete (EBSCO host), PsyInfo, ProQuest, and Web of Science, and used different combinations of keywords, including: "international student", "international graduate", "post-study", "post-graduation", "work", "migrant", "immigrant", "sojourner", and "education". Fig. 1 provides an overview of each stage of the literature search. After the initial keyword search and exclusion of duplicate records, we found 4023 relevant pieces of literature. We then screened each document by its title and abstract, and 803 articles were identified as potentially relevant after this step. To further assess the relevance of these articles, the following screening questions were applied: (1) Does the article examine the international graduates working in the host country? (2) Does the article study international students who are going to/have recently graduated and are seeking employment in the host country? (3) Does the article examine international students' employability and/or employment outcomes in the host country? Only articles that met one or more of the above selection criteria were included. After reading abstracts and identifying additional secondary references, a total of 74 articles were assessed as being relevant to our focal topic and were retained for inclusion in the review. Dates of publication ranged between 2002 and 2020. Studies on the topic increased significantly after 2011, with 66 articles published between

¹ In accordance with recent reviews of the literature relating to the workforce integration of newcomers (e.g., Hajro et al., 2019; Lee et al., 2020), we apply a broad definition of this concept, and view it as the process of entering the workforce in a host country and engaging in economic activities through employment.

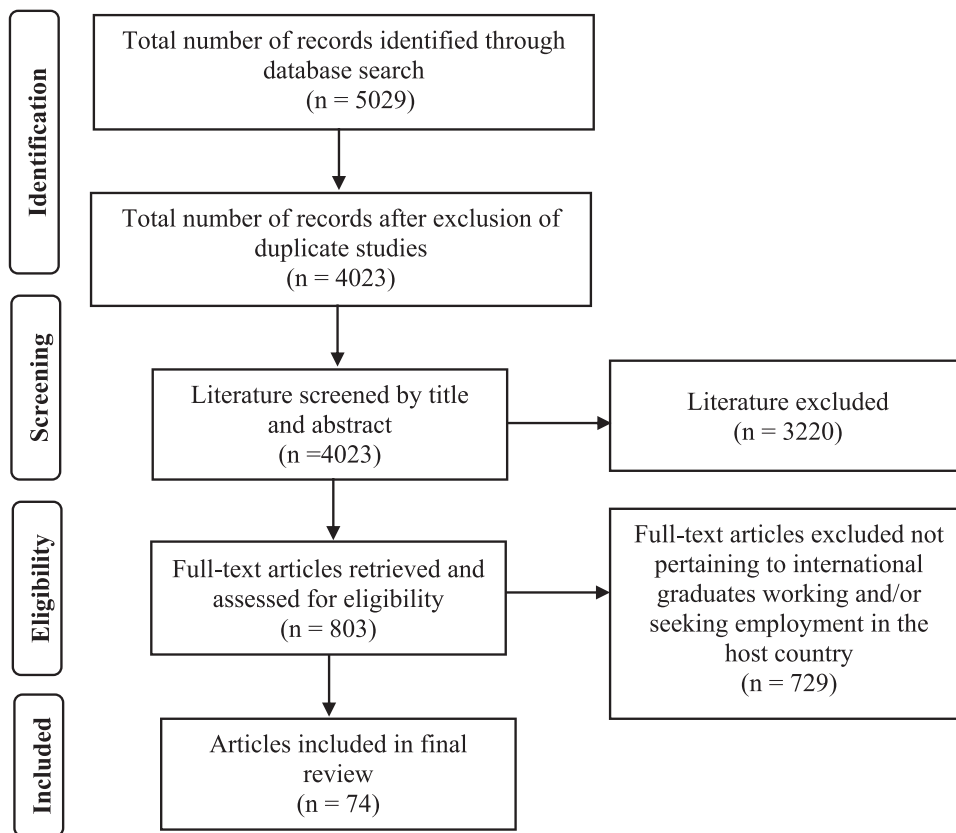


Fig. 1. PRISMA flow diagram of literature selection.

Adapted from Page, McKenzie, Bossuyt, Boutron, Hoffmann, and Mulrow (2021).

2011 and 2020. Table 1 provides an overview of the primary characteristics (e.g., discipline, nature of the sample, main variables studied, theoretical framework, research design) and findings of each of the articles included in the review.

We identified and organized the key findings and themes emerging from the papers into micro, meso and macro levels of analysis. These three levels of analysis have been used in previous reviews of the employment of globally mobile groups, such as expatriates and refugees (e.g., Doherty, 2013; Lee et al., 2020). Studies at the micro-level have examined how factors such as individual-level characteristics (e.g., demographic, cognitive, social, experiential) contribute to post-graduation employment while meso-level research has examined specific organizational attributes and practices that influence workforce integration. Macro-level influences relate to national and societal factors (e.g., government programs/policies, economic conditions) that contribute to international graduate work integration. Table 2 outlines the primary themes/factors that were found to influence international graduates' integration at each level, which we now discuss further.

Micro-level: Individual Factors

Research at the micro-level has identified a number of key variables that contribute to international graduates' work integration, including various demographic, social, cognitive, and experiential factors. Social Cognitive Career Theory (SCCT; Lent et al., 1994) provides a useful framework for understanding how various individual and external factors influence career choices and outcomes. In SCCT, personal inputs, including various individual attributes (e.g., demographic characteristics, areas of knowledge/skill, personality), and perceptions of external contingencies (e.g., family pressure, labour market) can directly and indirectly impact individuals' career choices and outcomes. Below, we discuss various micro-level predictors of international graduates' post-graduation work experiences, including motivational factors, demographic characteristics (e.g., gender, age, family status), areas of knowledge/skill (e.g., language fluency; cultural knowledge), and other social and experiential factors (e.g., social network, work experience, educational background).

Motivation

Various motivational factors have been shown to affect whether international students decide to stay and work in the host country. Drawing on SCCT, Bozionelos, Bozionelos, Kostopoulos, Shyong, Baruch, and Zhou (2015) found in their study of international

Table 1
Overview of literature.

Authors	Discipline	Sample	Country of study	Dependent variables	Theoretical framework	Research design	Key findings
Anderson et al., 2014	Education	Asian-born business graduates, phase one interview n = 40, survey n = 131; phase two interview n = 31, survey n = 45; phase three interview n = 27, survey n = 14	NZ	Post-graduation transition pathways and barriers	Engaged pedagogy (Madge et al., 2009); human capabilities framework (Walker, 2008)	Interview, survey	Two “functional capabilities”: having economic opportunities and developing a sense of affiliation, were related to post-study experiences.
Anisef et al., 2003	Ethnic and migration studies	University graduates, n = 71,422	Canada	Earnings	Social inequality perspective	Secondary data (census)	Earnings of immigrants, particularly those who came to Canada at an older age with foreign degrees, are not commensurate with their education level.
Arbeit & Warren, 2013	Sociology	College-educated immigrants, n = 18,361	US	Employment outcomes and wages	Human capital theory (Becker, 1962; Frazis, 2002); credential theory (Collins, 1971)	Secondary data (census, longitudinal survey)	There were labour market penalties for immigrants holding a foreign degree.
Arkoudis et al., 2009	Education	International final-year and recent graduates, n = 40; local students and recent graduates, n = 18	Australia	Employment outcomes and performance		Secondary data (Census, survey), interview	International student graduates have poorer employment outcomes than their local counterparts, especially if international graduates have lower levels of English language proficiency.
Arthur & Flynn, 2011	Education	International students graduating, n = 19	Canada	Career decisions	Career planning and three phases of the transition process (Arthur, 2007; 2008)	Interview	Barriers to migration included the fear of not securing employment and cultural and linguistic barriers.
Arthur & Flynn, 2013	Management	Graduate international students, n = 14	Canada	Career transition experiences		Interview	Difficulties and barriers in transitioning from students to employees may include a lack of experience, visa status, and a lack of network.
Baas, 2019	Migration		Asia-Pacific region	Education-migration industry	Two-step migration	Special issue introduction	There is an emerging “industry” that facilitates study-abroad trajectories and acts as broker for two-step migration pathways.
Blackmore & Rahimi, 2019	Education	International students graduating, n = 220	Australia	Employment outcomes	Doxa and capital (Bourdieu, 1990)	Interview	Visa eligibility and English language ability filtered out many international student candidates. Recruitment practices showed bias against international graduates.
Blackmore et al., 2014	Education	International students, n = 107	Australia	Post-study employment		Longitudinal interview	Visa status, poor communication skills, inadequate soft skills, and lack of local work experience were main barriers to employment for international graduates.

(continued on next page)

Table 1 (continued)

Authors	Discipline	Sample	Country of study	Dependent variables	Theoretical framework	Research design	Key findings
Blackmore et al., 2017	Education	Chinese accounting graduates, n = 13	Australia	Experience of transition from an Australian university into the Australian labour market	Human capital theory (Bourdieu, 1977); cultural capital (Bourdieu, 1986)	Interview	Recruitment processes required the display of cultural capital. International students with high levels of cultural and economic capital were able to convert this into social capital to their advantage in the labour market.
Boyd & Tian, 2018	Migration	STEM immigrants, n = 290,975; Canadian-born, n = 415,365	Canada	Employment in STEM occupations and earnings	Human capital theory (Becker, 1962)	Survey (Secondary data)	Regardless of the location of their STEM education, immigrants earn less than Canadian-born who received university degrees in Canada.
Bozionelos et al., 2015	Management	International graduate students, n = 139	UK	Perceptions and interest in international careers	Social Cognitive Career Theory (Lent et al., 1994, 1996, 2000; Lent et al., 2005)	Survey	Individual factors (self-efficacy with respect to working abroad and outcome expectancy) and perceived constraints were directly related to interest in an international career.
Buddelmeyer et al., 2013	Public administration	Visa applicants survey, n = 17,765	Australia	Labour market outcomes		Census, survey	Labour market engagement was positively correlated with the time of arrival in Australia, English proficiency, and Australian qualification.
Cameron et al., 2019	Education	International graduates, n = 1159	Australia	Employment intentions and outcomes	Human capital theory (OECD, 2001)	Survey, forced choice and open ended questions. Fuzzy-set qualitative comparative analysis	Work integrated learning, work experience, visa status, and discrimination influenced transitions to employment.
Carlson & McHenry, 2006	Psychology	HR specialist, n = 60	US	Employability		Experiment	Speakers with perceived accents or dialects were given a low employability rating.
Chen & Skuterud, 2018	Public administration	Former international students, n = 2971; foreign-born and educated immigrants, n = 19,361; Canadian-born and -educated graduates, n = 87,387	Canada	Labour market performance	Human capital theory	Secondary data	Former international students outperformed foreign-born and-educated immigrants but underperformed relative to Canadian-born and -educated graduates from similar post-secondary programs.
Chiou, 2017	Public administration	Two countries	Australia, NZ	Similarities and differences between the two countries two-step migration policies	Policy system (Considine, 1994, 2005)	Case studies, policies review	Australia and New Zealand pursued different paths to arrive at a similar approach toward international students seeking transition to permanent skilled migration after study.
Chira et al., 2013	Sociology	International students and graduates, n = 35; policy and service providers, n = 15	Canada	Decisions to study, work and settle in Atlantic provinces		Interview narrative analysis	Students encounter obstacles such as lack of networks and discrimination in entering the Atlantic Canadian labour market post-graduation.

(continued on next page)

Table 1 (continued)

Authors	Discipline	Sample	Country of study	Dependent variables	Theoretical framework	Research design	Key findings
Coffey et al., 2018	Management	International graduates, n = 874	Australia	Job searching barriers and experience	Human capital theory (Becker, 1964; 1993)	Survey with closed and open-ended questions	The discriminatory employment experiences of international graduates demonstrated a lack of congruence between the concept of best fit in hiring practices and a culturally diverse labour market.
Conrad & Meyer-Ohle, 2019a	Management	HR managers, n = 19; newly hired foreign graduates, n = 33	Japan	Foreign university graduates' employment	Varieties-of-Capitalism perspective (Hall & Soskice, 2001)	Interview	Foreign graduates viewed as resource for enhancing workforce and internationalizing Japanese firms.
Conrad & Meyer-Ohle, 2019b	Migration	Foreign young employees, n = 33; human resource sections, n = 19; brokers, n = 10	Japan	Recruitment of foreign workers	Two-step migration model	Interview	Japanese government has support for two-step student migration through its "Career Development Program for Foreign Students".
Crossman et al., 2020	Economics		Canada	Advantages and risks of two-step immigration selection process	Two-step migration (Hawthorne, 2010)	Conceptual	Immigrants who have successful previous experience in host country workplaces and educational institutions may have acquired relevant social network and interpersonal skills.
Dam et al., 2018	Migration		Canada	International students immigration after graduation		Secondary data	Canadian immigration system and policies created barriers for international students to become permanent residents.
Davis, 2017	Psychology	International graduates, n = 192	US	Stress level	Social cognitive career theory (Lent et al., 1994)	Survey, interview	There was a relationship between employment barriers and perceived level of stress for international students who pursue employment in the U.S. post-graduation.
Esses et al., 2018	Education	International student, n = 3424	Canada	Plans to work in Canada after graduation, intentions to apply for permanent residency		Survey	Demographic, economic, socio-cultural, and academic variables influence international students' plan to work in Canada and apply for permanent residence after graduation.
Farivar et al., 2019	Management	International graduates, n = 129	Australia	Change of career mobility intentions	Push–pull theory (Toren, 1976)	Survey, qualitative comparative analysis	Work environment was a strong condition for changing career mobility decisions.
Fogg & Harrington, 2012	Education	College graduates, n = 100,400	US	Earnings	Human capital theory (Becker, 1964)	Survey	Gender, marital status, region of residence in the US, country/region of college degree, English language proficiency, type of visa, level of education, work experience, major field of study may influence wages of immigrant workers.

(continued on next page)

Table 1 (continued)

Authors	Discipline	Sample	Country of study	Dependent variables	Theoretical framework	Research design	Key findings
Fossland, 2013	Geography	Recruitment Agencies, n = 9; skilled migrants, n = 27	Norway	Labour market participation	Economic, cultural, and social capital (Bourdieu, 1977; 1990)	Interview	Migrants' negotiations and employers' preferences influence migrants' employment.
Frank & Hou, 2018	Public administration	Immigrant workers, n = 5826; Canadian born workers, n = 24,985	Canada	Life satisfaction		Survey	Overeducation was negatively associated with life satisfaction, and the effect was weaker in the immigrant population than non-immigrants.
Geddie, 2013	Geography	Foreign students completing science and engineering graduate degrees, n = 47	Canada, UK	International graduates' employment and settlement considerations	Mobile subjectivities of migrants (Silvey & Lawson, 1999)	Interview	Relationship considerations are related to graduates' employment and settlement considerations at transitional life-stage.
Gribble, 2014	Education	International students	Australia	Work integrated learning, employability, employment outcomes		Literature review	Work integrated learning, employers' attitudes, English competency, and visa status can influence employment outcomes of international students.
Gribble et al., 2017	Education	International graduates transitioning into the Australian labour market, n = 13	Australia	The development of key employability skills	Personal capital (Brown et al., 2004)	Interview	Many of the barriers international students face when attempting to transition into the Australian workplace stem from their struggle to connect with the Australian society during their study experience.
Harrison & Ip, 2013	Education	International social work students	Australia	Fieldwork placement	Cultural capital (Bourdieu, 1986)	Conceptual, literature review	International social work students may be perceived to lack the requisite cultural capital to successfully complete their practicums.
Hawthorne, 2010	Migration	Skilled migrants	Australia	Labour market outcomes	Two-step migration	Secondary data	Former international students achieved comparable labour market participation rates to migrants recruited offshore.
Hawthorne, 2014	Migration	International students, n = 79,046; domestic students, n = 371,951		Study choices, engagement in two-step migration, and employment outcomes	Two-step migration	Secondary data	Australia's revised skilled migration requirements influenced international students' engagement in two-step migration.
Hawthorne, 2018	Migration	Three cases	UK, US, Australia	Evolving study-migration pathways	Economic theory of migration (Ruhs, 2008)	Case studies	Australian employers demonstrated an interest in the study-migration pathway. Their level of interest was influenced by occupational field.
Hou & Bonikowska, 2018	Migration	Immigrant women, n = 3.86 million person-years; men n = 4.09 million person-years	Canada	Earnings		Secondary data	Immigrants who were former temporary residents with a work or study permit had higher initial earnings than immigrants who first arrived in Canada as landed immigrants.

(continued on next page)

Table 1 (continued)

Authors	Discipline	Sample	Country of study	Dependent variables	Theoretical framework	Research design	Key findings
Hou & Lu, 2017	Sociology	Canadian-educated immigrants, n = 18,479; foreign-educated immigrants, n = 26,788; Canadian-born population, n = 28,517	Canada	Earnings		Secondary data	Canadian-educated immigrants had lower earnings than the Canadian-born population but higher earnings than foreign-educated immigrants.
Huang & Turner, 2018	Education	Mainland Chinese students, n = 449	UK	Students' attitudes to their employability and the support they engage with through their UK University	Processual approach of employability (Pegg, Waldock, Hendy-Isaac & Lawton, 2012)	Survey	International students were more aware of those work-related activities embedded in the program and their willingness to engage with central career services was limited.
Jiang & Kim, 2019	Education	International master's recipients, n = 1664; domestic master's recipients, n = 9972	US	Career outcomes (major and job match, annual earnings, and career satisfaction)	Human capital theory (Becker, 2009; Paulsen, 2001); Neo-racism theory (Balibar, 1992; Hervik, 2004)	Survey	International graduates were more likely than domestic master's recipients to hold jobs that are related to their master's degree. International master's recipients faced an earnings disadvantage.
Joseph, 2016	Sociology	Indian students gained permanent residence, n = 18	NZ	Experience of finding employment and transitioning to permanent residence	Social capital (Coleman, 1988; Portes, 1998)	Interview	Social networks facilitate international students' opportunities for settlement and providing support in their transition from temporary to permanent settlers.
Kim & Han, 2017	Management	International graduates, n = 499	US	International graduate students' intentions to remain in the US after completing their education.	Planned behavior theory (Ajzen, 1991, 2011); social cognition theory (Bandura, 2001; Bratman, 1999)	Survey	Professor support was related to the students' intentions to remain in the US. The relationship with the home country family and the perception of the home country labour market are associated with international students' intentions to remain.
Kim, 2015	Management	International graduates, n = 451	US	International graduate students' intentions to remain in the USA after graduation	Theory of planned behavior (Ajzen, 1991); role theory (Biddle, 1986)	Survey	There was a negative relationship between home country family ties and intention to remain for single students. The relationship between satisfaction with the university, social support and intention to remain was positive only for single students.
Kim, 2016	Sociology	Korean professionals graduated from US universities, n = 80	US	Inclusion and exclusion in global positional competition.	Global cultural capital and global positional competition (Brown, 2000)	Interview	Korean workers in US corporations have weaknesses in language and cultural skills.
Lawrence, 2013	Public administration	Interview with employers in Australia, Asia and the UK, n = 21; survey with international students, n = 2163	Australia, Asia and the UK	Employment outcomes of international students		Interview and online survey	Australian universities need to collaborate, embed employability skills within the curriculum, create alumni networks, and to provide more holistic approaches to internships.

(continued on next page)

Table 1 (continued)

Authors	Discipline	Sample	Country of study	Dependent variables	Theoretical framework	Research design	Key findings
Lu & Hou, 2015	Public administration	International students, 1990–2013 n = 1092,000	Canada	The rate of international students' transition into permanent residence		Secondary data	International students from countries with a lower GDP per capita had higher transition rates into permanent residence.
Lu & Hou, 2019	Public administration	Immigrants in Canada, n = 370,045; immigrants in the US, n = 223,608	Canada and the US	Mismatch between the education and occupations of immigrants		Census	In Canada, the over-education rate was lower among immigrants who were admitted through some form of employer selection than those who were admitted directly from abroad.
Lu et al., 2009	Migration	Mainland Chinese students, n = 160	Canada	Migration intentions	Push–pull theory (Toren, 1976)	Survey	Students' demographic characteristics, traits, Canadian experiences, parental expectations, and Canadian immigration policy influence students' migration intention.
Maury, 2017	Migration	African students have worked in Finland, n = 7	Finland	Migrants' experiences of temporal limits in life and at work	Critical migration theory (De Genova, 2013; Tazzioli, 2015)	Interview	Student-migrant workers have experiences of a fragmented lived time because of the temporary character of their visa.
Maury, 2020	Sociology, Economics	Non-EU/EEA student-migrants, n = 33	Finland	The experiences of non-EU/EEA student-migrants orienting in precarious labour markets in Finland	Contemporary capitalism (Dowling & Harvie, 2014; Harvey, 2017)	Interview	The exploitation of legally constrained working student-migrants is facilitated through the imposition of unpaid work hours in precarious employment.
Mosneaga & Winther, 2013	Geography	International students and graduates, n = 43	Denmark	International students' study-to-work transition process	Economic migration Model (Papademetriou et al., 2009); Complex patterns of skilled mobility (Favell et al., 2006);	Interview	Multiple personal and contextual factors can shape foreign students' perspectives on post-graduation careers in the host country.
Musumba et al., 2011	Economics	International graduate students, n = 470	US	Factors influencing career location preferences of international graduate students	Model of global and local knowledge (Lien, 2005)	Survey	Students prefer to start their careers in the country where they have more and better career opportunities, receive higher salaries, and have increased civil liberties.
Nourpanah, 2019	Migration	International students-turned migrants, n = 15	Canada	Experiences of international students enrolled in vocational nursing programs, subsequently transition to permanent residence	Normative and ideological dimensions of education	Ethnographic interview	Vocational courses have cultural, ideological, and political-economic significance to further migration projects at individual, regional, and national levels.
Nunes & Arthur, 2013	Psychology	International students entering the workforce, n = 16	Canada	Helpful and hindering factors in the transition from university to employment		Interview	Employers discriminated against students' international status, lack of citizenship, or poor English proficiency.
Oreopoulos, 2011	Economics	Resumes, n = 12,910	Canada	Number of call-backs received	Statistical discrimination (Phelps, 1972)	Field experiment	There is substantial discrimination towards job applicants with foreign experience or names.

(continued on next page)

Table 1 (continued)

Authors	Discipline	Sample	Country of study	Dependent variables	Theoretical framework	Research design	Key findings
Peng, 2020	Sociology	Mainland Chinese students to migrants, n = 24	HK	School-to-work transition process	Processual perspective	Interview	Transition of migrant students is an extended and ongoing process covering job-hunting, initial adaptation to working life, continued career development, and future plans. STEM immigrants educated in Canada, the United States, the United Kingdom or France had outcomes similar to the Canadian-born.
Picot & Hou, 2019	Public administration	Immigrants n = 135,780; Canadian-born individuals, n = 141,900	Canada	Skill utilization and earnings of employed STEM-educated immigrants		Census	
Popadiuk & Arthur, 2014	Management	International student, n = 18	Canada	International student university-to-work transitions	Relational-cultural theory (Jordan, Kaplan, Miller, Stiver & Surrey, 1991; Miller & Stiver, 1997)	Interview	Relationships support international students to successfully navigate the transition from university to work.
Rajani et al., 2018	Ethnic and migration studies	Indian graduate, n = 1	Canada	Decision processes of international students contemplating whether to stay or return home after completing their studies	Push-pull theory (Toren, 1976)	Autoethnographic account	Personal ambitions and positive experiences contributed to the student's adjustment and success as a self-initiated expatriate in the host country.
Robertson & Runganaikaloo, 2013	Ethnic and migration studies	Student-migrant, n = 35	Australia	Migrant experience within the education-migration nexus		Narrative analyses	Uncertainty and precariousness inherent in the student-to-migrant process create significant tensions and strategic responses in the daily lives of most student-migrants
Robertson, 2011	Sociology	Permanent residents who were international students, n = 20	Australia	Process of permanent residence acquisition	Migrant transnationalism (Yeoh et al., 2003)	Interview, research diary	International graduates experience the bureaucratic processes of application for permanent residence.
Robertson et al., 2011	Sociology	Singaporeans who gained permanent residency after completing study, n = 25 Singaporeans returning home from their studies, n = 24	Australia	Graduates' post study lives and careers		Interview	After graduation pathways were influenced by students engagement with international education as a life as well as a learning experience.
Sangganjanavanich et al., 2011	Psychology	International students seeking employment, n = 8	US	International students' employment search perceptions and experiences		Journal entries, focus group	International students experience both benefits and challenges related to their experiences in the United States.
Shchegolev et al., 2016	Management	Company recruiters, n = 12	Germany, Russia	Employment opportunities and barriers for international graduates with host country degrees	Push-pull theory (Toren, 1976); Human capital theory (Becker, 1964; Schultz, 1961)	Interview	Country-specific human capital components are highly valued by employers
She & Wotherspoon, 2013	Sociology	3 countries	Canada, US, UK	International student migration	Political economy	Secondary-data comparisons	International student policies, entry and immigration regulations shape the environment of student migration and the trends in student mobility.

(continued on next page)

Table 1 (continued)

Authors	Discipline	Sample	Country of study	Dependent variables	Theoretical framework	Research design	Key findings
Shen & Herr, 2004	Psychology	International students, n = 18; naturalized status student, n = 1; faculty members, n = 3; career counselors, n = 2	US	Career placement concerns of international graduate students	Vocational development theory (Super, 1969)	Interview	International students have negative perceptions of campus career center (e.g., limited services).
Suter & Jandl, 2008	Public administration		Australia, Canada, Scotland, UK	Settlement of foreign graduates		Policy analyses	Admission and retention policies influence settlement of foreign students.
Sweetman & McBride, 2004	Public administration	Canadian educated immigrant, n = 178,254	Canada	Labour market outcomes		Census	Postsecondary field of study is an important determinant of earnings for both the immigrant populations.
Tharenou, 2015	Management	Chinese foreign graduates		Mobility paths of Chinese international graduates	Push–pull theory (Toren, 1976); career capital theory (DeFillippi & Arthur, 1994)	Review	Chinese foreign business graduates may be caught in a dilemma when selecting the best path for their career and lifestyle. The stereotyping of International students as mere “migration hunters” impact their study and work experiences.
Tran & Vu, 2016	Education	International students, n = 105; staff members in vocational education and training institute, n = 50	Australia	Effects of stereotyping on international student study and work experiences	Positioning theory	Interview	The stereotyping of International students as mere “migration hunters” impact their study and work experiences.
Tran et al., 2020	Sociology	International graduates, n = 32; university staff, employers, agents, government representatives, n = 18	Australia	Effects of the post-study work policy in Australia	Two-step migration	Interview	The temporary graduate visa provides international students with a chance to improve their employability skills and social capital.
Wall et al., 2018	Education	International students in work integrated learning, n = 105	Australia	International students workplace learning experiences	Practice theory (Bourdieu, 1977)	Interview	Outlines challenges of international students participating in work integrated learning opportunities
Yang, 2017	Education, Sociology	Chinese students-turned migrants, n = 20	Singapore	Flexible citizenship	Cultural identity	Interview	Students turned migrants have a positive attitude toward mobility and flexibility.
Yu, 2017	Migration studies	Chinese graduate students during the transition period, n = 19	US	Social and cultural capitals and career experiences of international students	Human capital theory, social capital, cultural capital (Bourdieu, 1986)	Interview	Migrants’ social and cultural capital influence their career experiences at different stages of the transition.

Table 2

Multi-level framework of factors contributing to the workforce integration of international graduates, and avenues for future research.

	Factors influencing intl. graduates' work integration	Examples of future research questions
Micro-level	• Motivation • Country of education • Education level • Language fluency • Cultural knowledge • Social network • Work experience • Gender • Family status	Demographic characteristics • How do demographic characteristics of international graduates (e.g., gender, age, ethnicity) individually and jointly influence their workplace integration? Dispositional factors • To what extent do international graduates' dispositional characteristics (personality traits, values) affect their workplace integration? Are there specific characteristics that identify which international students are more or less likely to experience integration challenges? How do these characteristics interact with other micro-level factors (e.g., language fluency, social network, work experience) in influencing integration? Emotion • How do international graduates experience and manage their emotions during various phases of school-to-work transitions (e.g., job search, organizational entry, adaptation)? What strategies do international graduates use to cope with the gap between expectations and reality?
Meso-level	• Recruitment and selection practices • Workplace norms and expectations • Discrimination and harassment in the workplace • Post-secondary institution support	Organizational characteristics • What organizational characteristics (e.g., size, field, degree of internationalization, top management attitudes and beliefs) influence • International graduates' perceptions of their work experience and level of integration? Why do these effects exist? Organizational policies and practices • How can organizations improve HR/workplace practices (e.g., selection, onboarding, socialization) to enhance international graduates' workplace adaptation and integration? Discrimination • To what extent do various types of discrimination influence international graduates' work integration? • What are the specific employment barriers that international graduates face in organizations? • What policies/practices can be implemented to reduce discrimination/harassment against international graduates in the workplace? Educational institutions • What can educational institutions and educators do to collaborate with government agencies and organizations to construct programs and services that are better tailored to the needs of international students? • To what extent do different institutions, programs, and fields of study influence the effectiveness of different practices (e.g., co-op programs, career services) that can facilitate international graduates' integration?
Macro-level	• Immigration policy and work legislation • Economic conditions • Safety and stability of host country • Immigration industry	Laws and policies • What specific labour laws and policies are available and effective in ensuring the equal opportunities and workplace rights of international graduates? • What policies/regulations should be introduced (or removed) to better attract and retain international graduates? Country culture • What key cultural differences exist between the host country and the country of origin of international graduates? How do these cultural differences influence workplace integration in the host country? Government support • What kinds of government services and supports can be introduced or further developed to assist international graduates in their transition and adaptation to the workplace?

graduate students in the United Kingdom that self-efficacy in working abroad and outcome expectancy influenced international graduates' interest in an international career. Furthermore, factors such as adjustment during graduate studies, and exposure and immersion in the host country's culture can indirectly influence international career interests through self-efficacy of working abroad. Higher levels of personal ambition may also influence whether foreign students choose to return to a host country after returning to work in one's home country (Rajani et al., 2018).

According to the theory of planned behavior (Ajzen, 1991, 2011), an interest in becoming an immigrant after studying in the host country will motivate students to prepare to work abroad (Cameron et al., 2019). In this respect, strong intentions to remain abroad facilitate migration. However, the ambiguity and precariousness (e.g., changes in immigration policy) inherent in the immigration process can also create significant uncertainty and anxiety in the lives of student migrants (Robertson & Runganaikaloo, 2014). Indeed, student migrants must develop and maintain a willingness to disconnect from their home country while often encountering uncertainty regarding factors such as their immigration and work status. A qualitative study of twenty Chinese students who became working professionals in Singapore found that they use the term "flexible citizens" describing themselves, which implies a level of ambiguity in their immigration status, but also a readiness to disconnect from their home country (Yang, 2017).

Country of education

Several studies have examined international graduates' educational background in relation to their work integration. Drawing on human capital theory, [Arbeit and Warren \(2013\)](#) argued that educational credentials are a key indicator of the knowledge, skills, training that employees can bring to the labour market. However, educational credentials and human capital may not transfer evenly across countries such that skills and knowledge acquired in one's post-secondary education in one country may not be valued equally in other countries (e.g., [Akresh, 2007, 2008](#); [Arbeit & Warren, 2013](#); [Zheng & Xie, 2004](#)). Moreover, from the perspective of credential theory ([Collins, 1971](#)), the skills and knowledge associated with educational attainment are less relevant because employers also use educational credentials to assess employees' "status cultures". For example, graduates from a prestigious liberal arts college may be perceived as having certain background experiences and value orientations ([Arbeit & Warren, 2013](#)). Furthermore, students from a more academically-oriented research university may have different values and beliefs than students graduating from a college with a more applied/vocational focus ([Collins, 1971](#)). When employers assess the educational background of a potential employee, they assess what their degree(s) represent in terms of their status, beliefs, and values and evaluate whether a potential employee will fit in and reflect the norms of the organization ([Arbeit & Warren, 2013](#); [Collins, 1971](#)). It is more difficult to assess graduates' skills and status cultures from foreign colleges; thus, employers place less value on foreign degrees earned outside of the host country ([Arbeit & Warren, 2013](#)).

International graduates with host country degrees may be more valued by employers as they bring human capital from their home country and also acquire additional capital relevant to the host country context ([Shchegolev et al., 2016](#)). In this respect, studies have suggested that employers tend to highly value domestic degrees, and international graduates with degrees earned in the host country tend to yield more favourable career outcomes than immigrants with degrees earned abroad ([Oreopoulos, 2011](#)). For example, studies have shown that immigrants who have graduated from a host country institution are more likely to find a job matched with their qualifications (e.g., [Arbeit & Warren, 2013](#); [Fossland, 2013](#); [Shirmohammadi et al., 2019](#)).

Studies have also shown that host country educational credentials significantly influence a graduates' employment earnings. For instance, [Hou and Lu \(2017\)](#) reported that Canadian-educated immigrants had higher earnings than foreign-educated immigrants both in the short run and in the long run. Similarly, based on a survey of post-secondary graduates from Canadian public post-secondary institutions, [Chen and Skuterud \(2018\)](#) found that former international students outperform foreign-born and -educated immigrants in terms of their hourly earnings after graduation. In general, Canadian educated immigrants appear to have far better labour market outcomes, including hourly and weekly earnings, than foreign-educated immigrants ([Sweetman & McBride, 2004](#)). Similar results have also been found in other countries. In the United States, for example, the mean hourly wage of immigrants with a US college degree is higher than immigrants with college degrees from other countries, except for Canada, the United Kingdom, and Ireland ([Arbeit & Warren, 2013](#); [Fogg & Harrington, 2012](#)).

Given that immigration policies for various countries often prioritize certain skills sets and educational credentials in some disciplines and occupations, examining the transferability of one's education and skills has become an important issue. Science, technology, engineering, and mathematics (STEM) skills are highly valued by North American countries' immigrant policies. Thus, studies on STEM-educated graduates have recently gained attention in the literature. In their study of STEM-educated immigrants in Canada, [Picot & Hou \(2019\)](#) observed that country of education is a critical factor for STEM immigrants, as STEM immigrants educated in Canada, the United States, the United Kingdom, or France had employment outcomes (i.e., annual earnings: wages/salaries and net self-employment income) comparable to those of Canadian-born STEM workers. Similarly, [Boyd and Tian \(2018\)](#) reported that although immigrants educated in Canada, the United States, the United Kingdom, and France have slightly lower earnings than that of Canadian-born and educated workers, the earnings gap was much larger for immigrants educated elsewhere (i.e., Eastern Europe and Asia).

Education level

In addition to the location of one's education, education level is also associated with measures of international graduates' work integration. For example, in their analyses of the National Survey of College Graduates in the United States, [Fogg and Harrington \(2012\)](#) found that the earnings of immigrant college graduates increased with their education level. The hourly wage premium of a master's degree relative to a bachelor's degree was 16% among employed immigrants. Relative to a bachelor's degree, the premium of a doctoral degree was 32% ([Fogg & Harrington, 2012](#)).

Related to education level, over-education (i.e., when a worker's educational credentials significantly exceed those required for their job; [Hartog, 2000](#); [Rubb, 2003](#)) is a problem prevalent among immigrant workers ([Chiswick & Miller, 2009](#); [OECD/European Union, 2019](#)). Over-education, however, appears to be less severe for immigrants who were educated in the host country. For example, [Anisef et al. \(2003\)](#) found in a Canadian sample that visible minority immigrants' earnings were less likely to be commensurate with their education level only when they did not have Canadian educational credentials. Interestingly, although there is a greater incidence of over-education among immigrants, there is some evidence to suggest that the negative effect of over-education on life satisfaction is weaker for the immigrant population than for the non-immigrant population ([Frank & Hou, 2018](#)). Further research is needed examining factors that influence overeducation, including the discounting of foreign educational credentials.

Language fluency

In the literature on international graduates, language competence has been identified as one of the most critical factors associated

with successful work integration. International students' experiences studying abroad can significantly enhance their foreign language acquisition especially when host country languages are close to their native language (Sorrenti, 2017). Fluency in English has attracted the most attention in the literature as English is the first language of several countries with higher proportions of international graduates (e.g., Arkoudis et al., 2009; Blackmore et al., 2017; Gribble, 2014); however, other first languages have also been examined. For example, proficiency in Danish was found to help international graduates to expand their social networks in Denmark (Mosneaga & Winther, 2013).

In their study of international graduates pursuing permanent immigration to Canada, Arthur and Flynn (2011) found that English proficiency is one of the primary barriers to their successful integration (Arthur & Flynn, 2011). As foreigners in Canada, participants believed that they were able to communicate effectively in general in English; however, they encountered some difficulty in understanding some expressions and the use of nuanced language. For example, international graduates found it difficult to understand some idioms and jokes, and felt that this lack of comprehension may lower their colleagues' perceptions of their fit with the culture of their organization.

Among the dimensions of language ability, speaking ability has been emphasized as a key predictor of international graduates' career outcomes. Fogg and Harrington (2012) reported that spoken-English ability is positively related to the hourly earnings of international graduates in the United States. Likewise, Arthur and Flynn (2013) reported that international graduates in Canada are concerned that their fluency and accent compromise their employment prospects. Furthermore, in a study of various employers in Canada, Carlson and McHenry (2006) found that human resource specialists were more likely to assign a lower employability rating to job applicants with a strong accent in spoken English. Similar findings have emerged in other countries. For example, Peng (2020) reported that mainland Chinese migrants in Hong Kong face language barriers due to their lack of fluency in spoken Cantonese.

Cultural knowledge

Several studies have suggested that cultural knowledge, including greater familiarity with the values, attitudes, and behavioral norms that exist in the host country can enhance an international graduates' work integration. Arthur and Flynn (2011, 2013) found that specific cultural barriers (e.g., difficulties in understanding cultural expressions; ambiguous networking processes), and differences between one's home country and Canadian culture, can influence international graduates' perceptions of integration into Canadian society and the workplace. Participants in this research recommended that international students should learn more about the culture and norms of workplaces in the host country and the expectations of other employees (Arthur & Flynn, 2013). Likewise, Shchegolev et al. (2016) reported that recruiters in Russia and Germany appreciate and actively consider international graduates' familiarity with the host country culture and workplace norms in assessing potential recruits.

If international graduates lack knowledge of the social customs in the host country and in the workplace, they often experience uncertainty and anxiety (Fritz et al., 2008). In a qualitative study of international students graduating and seeking jobs in the United States, Sangganjanavanich et al. (2011) observed that greater perceived cultural differences resulted in greater frustration and discomfort, which hindered international graduates' success in searching for jobs. Similarly, Kim (2016) found that even though they possessed strong technical and mathematical skills, Korean workers employed in U.S. firms acknowledged that they experienced high levels of stress adapting to their workplace due to a lack of cultural skills (Kim, 2016).

Social network

In line with social and career capital theories (Blackmore et al., 2017; Bourdieu, 1986; DeFillippi & Arthur, 1994; Tharenou, 2015), a number of studies have highlighted the importance of international graduates' social and professional networks in securing suitable employment and developing their careers in the host country. In their study of international graduates in New Zealand, Joseph (2016) found that social networks play a key role in providing employment opportunities and facilitating international students' transition from temporary to permanent settlers. Similarly, Gribble et al. (2017) reported that having a local network either on campus or in the local community is critical for international graduates in their search for employment in Australia, and often form a significant barrier to their labour market entry.

Despite these findings, one of the key advantages that international graduates possess relative to other migrant groups is an opportunity to network and build relationships with other people on campus (fellow students, professors) and also in the local community before seeking employment. Support from fellow students and one's professors can help in identifying relevant job opportunities and securing employment (Kim, 2015; Kim & Han, 2017). Some professors/supervisors can provide international students with information on jobs that are available and also introduce them to key company contacts (Arthur & Flynn, 2011; Popadiuk & Arthur, 2014). This support helps reduce barriers in the job application process and provide more direct links to employers.

Building networks with fellow students from the same home country can also assist international graduates in their transition to working in the host country. In a recent study of mainland Chinese students working in Hong Kong, some graduates reported that they primarily engaged in social activities with other students from the Chinese mainland (Peng, 2020). Although these students enjoyed interacting with local work colleagues, due to cultural differences and their heavy workload, they found it easier to maintain ties and social connections with others from the Chinese mainland than local individuals. Similarly, Yu (2017) found in a qualitative study of a group of Chinese migrants in Singapore that students-turned-migrants maintained close interpersonal connections and an emotional attachment to their home country through the use of information technology (i.e., social media).

Despite this evidence that international graduates seek to retain social and emotional connections to their home country, some studies have also reinforced the importance of developing a social network that includes local nationals. For example, in their

qualitative study of international students in Denmark, [Mosneaga and Winther \(2013\)](#) found that most participants sought to create a locally rooted social network, and one that does not resemble an “international bubble” in order to feel more integrated into the host country ([Mosneaga & Winther, 2013](#)).

Work experience

Prior work experience, particularly in the host country, can also be instrumental in enhancing international graduates' work outcomes. In their study of international students transitioning from study to work in Canada, [Nunes and Arthur \(2013\)](#) observed that a lack of work experience was a primary barrier in the job search process for unemployed graduates. Because immigrants who have previous experience working in the host country are able to expand their social networks and acquire skills that enhance their employability (e.g., communicating effectively, working in a team), prior work experience in the host country can be a critical asset ([Crossman et al., 2020](#)).

Consistent with this premise, [Hou and Bonikowska \(2018\)](#) found that, controlling for other factors (e.g., age, education, years of residence in Canada), immigrants who were former temporary residents with prior Canadian work experience had much higher initial earnings than immigrants without such experience. Furthermore, a key factor differentiating Canadian-educated immigrants' post-immigration earnings from the earnings of Canadian-born and foreign-educated immigrants is whether they held a well-paid job in Canada before becoming permanent residents ([Hou & Lu, 2017](#)). Although Canadian-educated immigrants have higher earnings than foreign-educated immigrants, Canadian-educated immigrants without any work history in Canada before becoming permanent residents earned less than foreign-educated immigrants who are equivalent in other areas ([Hou & Lu, 2017](#)). While host country work experience plays a pivotal role in international graduates' work outcomes, work experience in one's home country before studying abroad can also impact international graduates' work integration. If international graduates have limited or no previous work experience, this can significantly negatively impact their employment prospects and ability to find suitable work ([Coffey et al., 2018](#); [Gribble et al., 2017](#)).

Gender

In their conceptual model of skilled migrants' qualification-matched employment in a host country, [Shirmohammadi et al. \(2019\)](#) propose that gender can moderate the effects of various predictors of qualification-matched employment. For example, they posit that the positive effect of individual-level factors (e.g., host country language proficiency, host country educational credentials) on access to jobs that matched their qualifications can be weaker for migrant women than men ([Shirmohammadi et al., 2019](#)). Surprisingly few studies, however, have tested these propositions or the influence of gender on the work experiences and integration of international graduates. There is some evidence that gender may interact with other factors in shaping employment outcomes for international graduates. In their study examining the labour market outcomes of college-educated immigrants in the United States, [Arbeit and Warren \(2013\)](#) found that female migrants who earned their degrees in the United States are more likely to be employed and earn higher wages than migrant women with foreign degrees. Furthermore, [Geddie \(2013\)](#) argues that traditional gender roles may influence whether and how effectively male and female migrants integrate into a host country's labour market. For example, some female migrants who are married and have children may choose to focus on family responsibilities rather than use their personal resources (e.g., education, work skills, cross-cultural experiences) to seek employment and advance their careers in the host country ([Fossland, 2013](#)). Taken together, this research suggests that female international graduates may experience different work outcomes than male graduates; however, this issue warrants greater research attention.

Family status

Family status (e.g., marital status, family ties in host/home country) can also play a significant role in shaping international graduates' employment and career outcomes ([Geddie, 2013](#)). The location of one's parents, for example, has been shown to influence where one chooses to work, with some international graduates being reluctant to settle in the host country because they want to be close to their parents ([Geddie, 2013](#)). Likewise, in an autoethnographic account of an Indian student in Canada, [Rajani et al. \(2018\)](#) explored the factors that influenced the student's experience studying in Canada, returning home, and then settling in Canada. The primary factor contributing to the decision not to stay in Canada in the first place were family ties, especially parents in the home country. In a qualitative study of international students completing science and engineering graduate programs in the United Kingdom and Canada, [Geddie \(2013\)](#) found that providing care for aging parents and/or children, managing dual careers, and work-life balance were key issues that influenced international graduates' employment and settlement considerations in the transition from school to work.

Related to these considerations, marital status is a significant predictor of an international graduate's intentions to stay to work in the host country. In their study of 451 international graduates in the United States, [Kim \(2015\)](#) reported that married participants with their partners in the U.S. had a greater propensity to stay longer. A similar pattern was observed by [Kim and Han \(2017\)](#), who found that Indian students had a greater propensity return to their home country than Chinese or Korean students as Indian students were more likely to be single. Interestingly, some studies have also suggested that marital status may moderate the effects of other predictors on students' career outcomes. [Fogg and Harrington \(2012\)](#) uncovered evidence of a “marriage premium” for male international graduates in the United States, such that married male college-educated immigrants earned considerably higher hourly wages than men who are not married. Likewise, [Kim \(2015\)](#) found that the positive influence of social support, and the negative influence of home country family ties on intentions to remain in the United States are only observed for single students ([Kim, 2015](#)). Taken together, these

studies highlight the importance of one's marital status and family ties in contributing to settlement decisions and where one pursues their career.

Meso-level: Organizational Factors

As outlined in Table 2, several key meso-level factors have been examined in the literature relating to international graduates' workforce integration, including employer recruitment and selection practices, workplace norms and expectations, discrimination and harassment in the workplace, and post-secondary institution support.

Recruitment and selection practices

Among various employer practices that influence the work integration of international graduates, recruitment and selection practices are the most frequently studied. Employers' perceptions and assessments of the overall capabilities, and the knowledge, skills, and abilities of international students are significant determinants of their participation in work placements and successful post-study employment (Gribble, 2014). Some studies have indicated that employers are reluctant to take on international students due to concerns over their English language skills and their capacity to fit into the workplace (Gribble, 2014). International students often feel that employers view them as less desirable than domestic students (Anderson et al., 2014; Arthur & Flynn, 2011). Indeed, in Australia, there is some evidence that employers favor domestic over international students in their hiring practices and in the salaries offered (Coffey et al., 2018; Hawthorne, 2018). Foreign students with an Australian degree are half as likely to be employed in a professional role than their Australian born peers (Buddelmeyer, Van de Ven, & Zakirova, 2013). This trend is particularly evident in occupations that are more volatile and in which applicants are in greater supply (e.g., accounting, business, engineering; Hawthorne, 2018).

In addition to concerns regarding the language and work-related skills of international graduates, employers also have concerns over their visa status. Employers tend to believe that the process of hiring an international graduate is expensive, lengthy, complicated, and risky (Blackmore, Gribble, Farrell, Rahimi, Arber, & Devlin, 2014; Gribble et al., 2017). Employers are therefore reluctant to hire international graduates because they fear they may not be able to secure the required work permit and may be forced to leave the country. Small firms, in particular, may not be able to afford to sponsor work visas for students (Rajani et al., 2018). Although specific post-graduation work rights and regulations have been introduced in some countries to facilitate this process, there is still evidence that employers prefer permanent residents and citizens over international graduates holding limited-length work permits (e.g., Arthur & Flynn, 2013; Tran, Rahimi, Tan, Dang, & Le, 2020).

Although several studies indicate that international graduates face negative perceptions and challenges in the hiring process, others have highlighted the unique strengths of international graduates. For example, Shchegolev et al. (2016) found that recruiters from companies in Germany and Russia highly value international graduates' intercultural competencies and their multilingualism. International graduates with a host country degree possess both home-country-specific as well as host-country-specific knowledge, which are both viewed as beneficial by employers, particularly if the company is planning to enter new foreign markets (Shchegolev et al., 2016).

Workplace norms and expectations

In addition to challenges faced in one's job search, international graduates may also encounter difficulty in adjusting to unfamiliar workplace norms and expectations. After starting a new job, graduates need to adapt to new schedules, work roles, and a different lifestyle from being a student. In their study of international graduates in Hong Kong, Peng (2020) found that several participants felt that the transition in lifestyle from school to work was difficult. The transition from school to work requires that graduates adapt to more intensive and often less flexible schedules, work role ambiguity, and exercise greater psychological and financial independence. The lack of structure and high levels of uncertainty in the workplace can present significant challenges to new graduates as employees are expected to demonstrate initiative in completing their work, sometimes without clear guidance/direction (Arthur & Flynn, 2013). Cultural differences may further magnify these challenges.

International graduates also face additional challenges in the workplace, including limited opportunities for career development, and in some cases, a lack of support from their organization in their efforts to adjust to the host culture. Wall et al. (2018) found in their sample of international students in Australia that participants felt that their previous work experience, knowledge, skills, and other capabilities were not recognized in the workplace, which limited their opportunities to engage in more challenging, highly skilled work tasks. Likewise, Conrad and Meyer-Ohle (2019a) found that international graduates in Japanese companies have difficulty understanding and adapting to Japanese HR practices that emphasize seniority in performance management and pay systems, unclear job descriptions, and generalist career paths. Complicating these efforts to adapt, some studies have suggested that employers are not providing sufficient support to assist international graduates in integrating effectively. For example, international graduates in Canada have noted that companies do not have sufficient knowledge of the immigration process, and they do not know how to sponsor international students to stay in Canada (Arthur & Flynn, 2011).

Although international graduates may have trouble in adjusting to workplace norms and expectations in a host country, many international graduates still report a positive view of their work experiences overall – a key factor associated with intentions to stay in the host country. For example, Arthur and Flynn (2011) found that international graduates generally have positive perceptions of the work environment in Canada, particularly relating to the reasonable expectations of employers, feeling respected, and being paid for extra work (overtime). In their study investigating the effects of various external conditions on international graduates' career

mobility, Farivar et al. (2019) identified favorable views of the work environment as the strongest factor altering the initial intention of international graduates from leaving Australia to staying after graduation.

Discrimination and harassment in the workplace

Several studies underline that bias and discrimination by employers can limit international graduates' employment options and their ability to integrate effectively. In their hiring processes, employers often favor candidates who will "best fit" with the culture of the organization (Blackmore & Rahimi, 2019). If organizational values and norms are not culturally inclusive, this can lead to discriminatory hiring practices that place international graduates at a distinct disadvantage relative to local applicants (Blackmore & Rahimi, 2019; Coffey et al., 2018). Several studies have uncovered evidence of discrimination in employers' hiring practices (e.g., Booth et al., 2010; Coffey et al., 2018). Studies have shown, for example, that immigrants who are from countries that are more culturally distant from the host country are more likely to be assessed by employers as less qualified and suitable job candidates (Dancygier & Laitin, 2014; Oreopoulos, 2011; Veit & Thijsen, 2019). In some cases, employers' discriminatory practices are intentional and not inadvertent. For example, in their study of employers in Japan, Conrad and Meyer-Ohle (2019a) found that although senior managers in Japanese firms were encouraging the recruitment of international graduates to assist in achieving diversity goals, front-line managers did not support this initiative. In response, HR managers adjusted selection criteria to increase requirements relating to language fluency and cultural fit. The use of these more conservative selection criteria shifted these companies away from their initial aims of increasing workforce diversity by hiring international graduates (Conrad & Meyer-Ohle, 2019a). Similarly, in Peng's (2020) study of international graduates in Hong Kong, some participants reported that Hong Kong employers and colleagues discriminated against them due to their mainland Chinese background. They reported experiencing verbal harassment during their interview and after being hired. In other cases, employers may not discriminate (or may not intend to discriminate) against international graduates; however, they are afraid that hiring too many foreign workers may negatively impact their business because their customers may be less accepting of foreign employees (Tran & Vu, 2016). Although employment discrimination exists in various organizations, some studies suggest that it is more prevalent in smaller firms as they are more likely to lack diversity and resources devoted to diversity management (Banerjee et al., 2018; Esses, 2021; Zschirnt & Ruedin, 2016).

Stereotypes of international graduates can fuel discriminatory behavior in the workplace. In their study of international students in Australian vocational education and training programs, Tran and Vu (2016) found that international students viewed themselves as being unwelcome in the workforce due to a prevailing stereotype depicting them as "permanent residence hunters". Some international graduates working in Australia also reported harassment in the form of informal jokes and humor referring to ethnicity and skin colour (Wall et al., 2018). Similarly, in their study of international graduates in Atlantic Canada, Chira et al. (2013) observed that some international graduates experienced discrimination due to their foreign accents. For instance, a bilingual (English and French) African student who graduated from a Canadian university noted she experienced discrimination during her work at a call center because she is a visible minority who speaks French from France, which sounds different than Canadian French (Chira et al., 2013).

Several studies suggest that international graduates may also experience pay discrimination. According to national survey data, Canadian-educated immigrants, on average, have much lower initial earnings than the Canadian-born population (Hou & Lu, 2017). In the Canadian labour market, former international students underperformed Canadian-born and Canadian-educated graduates from similar post-secondary programs (Chen & Skuterud, 2018). A recent study comparing international students' earnings with those of domestic students during their first five years after graduation from Canadian post-secondary institutions uncovered a significant earnings gap favoring domestic students (Choi, Hou, & Chan, 2021). This earnings gap is even larger at higher levels of education (i.e., comparing graduates with master's or doctoral degrees vs bachelor's or lower; Choi, Hou, & Chan, 2021). Likewise, Jiang and Kim (2019) found that international graduates employed in the United States who obtained their master's degree from American academic institutions faced an 8% earnings disadvantage compared with domestic graduates who also hold master's degrees.

In addition to receiving lower earnings after graduation, some international students also experienced lower levels of compensation for their work before graduation. According to Wall et al.'s (2018) survey of international students in work integrated learning programs (i.e., post-secondary education programs involving work placements, internships, practicum, project-based learning experiences, field trips, or voluntary work), many respondents felt that they were being exploited by employers. Some international students reported that they had to work such long hours that they equated it to being "unpaid" and felt that they were paid lower than other local employees with even fewer responsibilities (Wall et al., 2018). At the same time, however, international students were willing to accept unequal pay because they relied on the workplace learning opportunity to acquire relevant work experience, knowledge and skills in the host country. Taken together, this research indicates that international graduates may be subjected to discriminatory practices both in employers' hiring process as well as after securing employment.

Post-secondary institution support

In addition to employers, post-secondary institutions play a key role in international graduates' work integration. Work integrated learning and vocational programs can provide international students with a labour market advantage by developing transferable and marketable skills (Cameron et al., 2019; Wall et al., 2018; Wilton, 2012). In a study of international graduates from two Australian universities, work integrated learning has been identified as a key vehicle for improving graduates' employability (Cameron et al., 2019). In an ethnographic study in Canada, Nourpanah (2019) analyzed the experiences of international students who enrolled in vocational nursing programs then transitioned to temporary work upon graduation and became permanent residents. Participants noted that the vocational programs were worth the expense as they provided valuable work-related training and facilitated their

settlement and employment after graduation.

Although universities have increasingly invested in initiatives that enhance the employability of graduates in recent years (Artess, et al., 2017; Cranmer, 2006), international students often report that they receive limited attention and support from their post-secondary institutions in securing employment after graduation. In their study of mainland Chinese students in the United Kingdom, Huang and Turner (2018) found that students were aware of various curriculum-embedded activities that are designed to enhance student employability (e.g., work-based learning, course modules related to job search/seeking employment); however, they were less likely to engage with resources that enable more personalized career support/guidance (e.g., career services counselors, student societies). Likewise, Shen and Herr (2004) reported that students transitioning from studying to working in the United States expressed dissatisfaction with their career center due to the center's limited services and a perception that the services offered were not directly relevant to the graduating students' specialized fields (Shen & Herr, 2004). Similarly, in their study of international students graduating from schools in the United States, Sangganjanavanich et al. (2011) found that many participants believed that their career counselors did not understand their situation as international students and were not familiar with laws and regulations regarding the post-graduation employment of international workers. Taken together, these findings suggest that post-secondary institutions should explore ways of offering services and support that are better tailored to the unique needs and challenges faced by international students in seeking employment in the host country.

Macro-level: National and societal factors

According to social cognitive career theory, environmental contingencies, including factors such as the state of the economy, the policy environment, and various labour market characteristics, can influence individuals' perceptions of career barriers and opportunities (Lent et al., 2000). Accordingly, at the macro-level, various push and pull factors in the host and home country can influence international graduates' work integration, including economic conditions in both countries, host country immigration policies and work regulations, safety and stability, as well as the immigration industry.

Immigration policy and work legislation

Immigration policies directly affect whether international students can work and how long they can work in a host country after graduation. Many countries view international students as critical to building a highly skilled workforce and have implemented policies designed to retain them after graduation (Suter & Jandl, 2008). Canada and Australia, for example, have policies that facilitate international students' transition to skilled migrants and have invested in enhancing the school-to-work transition process (also referred to as "two-step migration" or the "education-migration nexus", Hawthorne, 2010, 2014; Robertson & Runganaikaloo, 2014; Yu, 2017). Similarly, the Japanese government has set a target of increasing the proportion of international students finding work in Japan after graduation from 30% to 50% and has initiated various programs (e.g., "the Career Development Program for Foreign Students") to prepare international students to work in Japan after graduation (Conrad & Meyer-Ohle, 2019b; Yonezawa & Yonezawa, 2016). Some countries have also developed regional immigration schemes that encourage international graduates to obtain employment and settle in certain parts of the country (Suter & Jandl, 2008).

Several studies suggest that perceptions surrounding work permits, visas, citizenship, and residency requirements can significantly influence international graduates' work integration. In a study of Chinese students in Canada, Lu et al. (2009) found that immigration policy changes designed to attract international students to stay (e.g., extending the length of the post-graduation work permit; providing off-campus work permits; the Canadian Experience Class immigration program) increased their intentions to stay in Canada after graduation. Providing post-graduation work permits appear to be a particularly effective means of improving the workforce integration of international graduates as they offer multiple benefits, including providing students the opportunity to improve their language proficiency, work-related skills, and to build their professional network before seeking employment (Tran et al., 2020). Likewise, Mosneaga and Winther (2013) found that non-EU/EEA graduates weighed various factors in their decision to stay in Denmark, with the stability of their residence status (e.g., length of temporary residency, eligibility for permanent residency) playing a key role.

Bureaucratic processes related to applying for residency and visas can be time-consuming and financially costly, and changes to these policies can increase applicants' levels of confusion and distress (Hawthorne, 2014; Robertson, 2011). In both Australia and Canada, students-turned-migrants must undergo a series of assessments (e.g., skills assessment, English test, health checks, and police checks) before they can apply for permanent residency (Robertson, 2011). In the United States, the H-1B visa (temporary work permit for foreign nationals with a bachelor's or higher degree) application process requires the employer to obtain Labor Conditions Approval from the Department of Labor and then file the Petition for a Nonimmigrant Worker (Form I-129) to the U.S. Citizenship and Immigration Services (USCIS). As more applications are received than the annual cap on new H-1B visa, USCIS conducts lottery-style draws to select H-1B petitions for further adjudication. International graduates seeking jobs in the United States experience stress and frustration due to the uncertainty involved in the application and the lottery process (Kim, 2016). Moreover, Maury (2017, 2020) argues that the temporary character of work visas and other elements of immigration policy heightens uncertainty in student-migrant workers' lives and puts them in a structurally unequal position in the labour market. Many countries (e.g., the United States) have student visa policies that restrict international students from working while enrolled in their program of study. Moreover, Gribble (2014) found that visa restrictions prevented international students' from participating in work-integrated learning programs (e.g., internships; co-op) in Australia. Compared to domestic students, international students in Australia have significantly lower participation rates in these programs (Gribble, 2014). Given that work experience before graduation is a key driver of employment after

graduation, these restrictions place international students at a significant disadvantage regarding their employability.

After graduation, if international graduates want to continue to work legally, they have to find ways to renew their work permit and/or attain permanent residency or citizenship before the expiration of their temporary work permit. In a study of student-migrants in Finland, some interviewees reported that they accepted very low-paid and precarious jobs in order to continue to live in Finland (Maury, 2017). Similarly, Dam et al. (2018) found that the Canadian Express Entry immigration processing system puts international students in precarious situations at risk of losing their ability to work and immigrate.

In addition to immigration and visa policy, other work-related policies and legislation can also influence international graduates' work experience. For example, Australia has established affirmative action and anti-discrimination legislation at both the Federal and State levels, such as the Racial Discrimination Act (1975), Australian Human Rights Commission Regulations (2019), and Victoria Equal Opportunity Act (2010). These laws promote equal employment opportunities and aim to protect minority groups from discrimination and harassment in the workplace. Similarly, Canada also has Employment Equity Act legislation designed to remove employment barriers for designated groups including visible minorities. At the same time, there is also legislation in some sectors that restricts employment options for immigrants. For example, the Public Service Employment Act (PSEA) in Canada specifies that preference in federal civil service positions hiring should be given to Canadian citizens (Wayland & Sheffield, 2014). Therefore, government-related employers have to give priority to Canadian citizens and permanent residents in their selection processes (Rajani et al., 2018). Moreover, for most jobs, if an employer intends to hire a foreigner applicant, they must establish that the position was advertised widely, provide clear evidence that they could not find a qualified Canadian or permanent resident for the position, and they must complete an annual summary report on their recruitment practices (IRCC, 2017). These administrative restrictions and procedures have the net effect of reducing employers' propensities to hire international graduates. Likewise, in the United Kingdom, Civil Service Nationality Rules (2021) require that non-reserved posts in the Civil Service are only open citizens of the United Kingdom, the Republic of Ireland, and the Commonwealth. These rules prevent international graduates without eligible citizenship from working in some public sector roles.

Economic conditions

Economic conditions both in the host and home country play a key role in determining where international graduates choose to work. International students are drawn to countries with more robust economies as they can provide stronger employment prospects and assist in achieving a higher quality of life (Carr et al., 2005; Kim & Han, 2017). Using survey data in the United States, Musumba et al. (2011) found that both economic and social factors influenced international graduate students' career preferences. When choosing between the United States and their home country, graduates preferred to work in the country with a higher standard of living, stronger civil liberties, better public safety, and cleaner water. Similarly, Mosneaga and Winther (2013) found that international students who continued to work in Denmark after graduation identified the quality of life and the welfare system as key benefits. Regional economic conditions can also influence settlement decisions and where international graduates choose to work. For example, in the United States, the hourly wages of immigrants who live in the Northeast and the West regions are higher than those who live in the South, which may partly explain generally higher immigration rates to these regions (Fogg & Harrington, 2012).

Related to economic conditions, when international graduates are from a home country with a weak economy and where they expect fewer job opportunities, they are more likely to remain in the host country (Kim & Han, 2017). Studies have found that high unemployment rates and less robust social and political systems in the home country increase international graduates' intention to work and settle abroad, and subsequently, are associated with higher levels of life satisfaction in the host country (e.g., Farivar et al., 2019; Tharenou, 2015). Using Canadian immigration databases, Lu and Hou (2015) confirmed that international students from countries with a lower GDP per capita tend to have higher transition rates into permanent residence than those who migrated from countries with a higher GDP per capita.

Safety and stability of host country

Related to host country economic conditions, host country safety and stability are also viewed as key considerations in international graduates' settlement decisions and work integration. In their interviews with international students in Canada, Arthur and Flynn (2011) found that participants had a positive view of Canada's legal system and felt the country offers a safe and stable place to live and work. Likewise, Esses et al. (2018) found that international students who perceive Canada as a safe place to live were more likely to plan to work in Canada and apply for permanent residency after graduation. Similarly, Musumba et al. (2011) found that international students were attracted to the United States due to its relatively high levels of political stability and public safety. These findings are consistent with research relating to other migrant groups. For example, Papademetriou et al. (2009) identified tolerance and safety in the host country as a key predictor of migrants' destination choice, especially for migrants with families.

Immigration industry

In addition to these broader societal factors, the immigration industry connecting international students to host countries has been attracting increasing attention in the research literature. The education-migration industry, which includes key actors such as: education agents, recruiters, money lenders, transnational companies, NGOs, and migrant associations, facilitates international students' study-abroad trajectories and acts as a broker for two-step migration pathways (Baas, 2019; Chiou, 2017; Hawthorne, 2010). For example, Conrad and Meyer-Ohle (2019b) focused on commercial brokerage and noted that brokers (international student recruiters)

helped change employers' attitudes regarding the employability of international students. Using the migration industry in the Asia-Pacific region as an example, Baas (2019) described how the increasing conflation between categories of international students and skilled migrants has led to the burgeoning education-migration industry (Khan, 2019; Sørensen & Gammeltoft-Hansen, 2012). One of the issues that has emerged in this industry is the presence of "bad actors", who have in some cases exploited international students through unscrupulous or fraudulent recruitment activities (e.g., charging exorbitant fees, advertising jobs/schools that do not exist; Altbach & Reisberg, 2013; Forde & MacKenzie, 2010; Huang et al., 2016). Given the expanding scope of international labour recruitment, government organizations and employers should work together to better identify and eliminate unscrupulous recruiters.

Discussion

We have provided a cross-disciplinary review of the literature regarding the workforce integration of international graduates. This review extends beyond scholarly reviews pertaining to other migrant groups (e.g., skilled migrants, refugees, international students; Hajro et al., 2019; Lee et al., 2020; Zhang & Goodson, 2011) by focusing specifically on international graduates and issues/themes relating to their work integration after graduation. Although research exploring the workforce integration of international graduates is a relatively new and expanding area, several key themes emerged relating to different levels of analysis. Below, we elaborate on these themes and outline several promising avenues for future research. Table 2 provide a summary of these proposed research directions.

Micro-level

At the micro-level, studies have emphasized the importance of various individual characteristics in contributing to the workforce integration of international graduates, including factors such as education level, language fluency, and previous work experience. Most studies included in the review, however, focus on a limited range of work and career outcomes, often focusing on objective measures of integration, such as salaries/income. In order to gain a more complete portrait of international graduates' post-graduation work experiences, we encourage researchers to assess a wider range of work and career outcomes. While studies assessing economic indicators of success are valuable in identifying certain barriers to successful integration (e.g., pay discrimination), further research is needed at the micro level that integrates relevant theories from the literature in the social sciences (e.g., cross-cultural psychology, sociology) and Management, and measures different facets of one's work experience, including the cognitive, affective, and behavioral components of these experiences. For example, additional research is needed assessing international graduates' perceptions of workplace inclusion and how these perceptions may be enhanced. Workplace inclusion refers to whether "people of all identities and many styles can be fully themselves while also contributing to the larger collective, as valued and full members" (Ferdman, 2017, p. 235). Shore, Randel, Chung, Dean, Ehrhart, and Singh (2011) have proposed that there are two key elements of workplace inclusion: belongingness and uniqueness. Thus, employers can strengthen their employees' perceptions of inclusion by fostering a sense of belonging (e.g., sharing information with employees, enabling their input/participation in decision-making) while also recognizing and supporting their uniqueness (e.g., respecting and supporting individuals' cultural differences and unique talents; Shore et al., 2018). Future studies should assess the perceived workplace inclusion of international graduates and factors that contribute to these perceptions.

Conducting more in-depth analyses of whether and how various individual characteristics (e.g., demographic, dispositional) influence work outcomes is another promising area for future research. Although demographic characteristics, such as gender, ethnicity/country of origin, and age can impact employee work experiences and objective measures of integration (e.g., earnings; Adamuti-Trache, & Sweet, 2005; Adsera & Chiswick, 2007; Cheung & Phillimore, 2017; Constant et al., 2009), these factors have not been examined in relation to different work and career outcomes, and their joint effects have been understudied. Indeed, while there is some evidence that female international graduates may experience different career outcomes than men (Anisef et al., 2003; Arbeit & Warren, 2013; Fogg & Harrington, 2012), few studies directly examine the influence of gender on international graduates' work experiences or how gender intersects with other demographic factors in shaping career outcomes. Furthermore, personality traits, such as the "Big Five" constructs, have been shown to predict a wide array of employee work outcomes, ranging from job satisfaction and work engagement to the quality of their relationships with their coworkers and leaders, to different components of their performance (Bruk-Lee, Khoury, Nixon, Goh & Spector, 2009; Chiaburu, Oh, Berry, Li & Gardner, 2011; Dulebohn, Bommer, Liden, Brouer & Ferris, 2012; Judge et al., 2002; Salgado, 2002; Young, Glerum, Wang & Joseph, 2018). Previous research has also signaled that individual differences relating to one's proactivity (i.e., proactive personality) can play an important role in shaping career outcomes, including one's career adaptability and success (e.g., Fuller & Marler, 2009; Jiang, 2017). Individuals who are proactive are more likely to expand their social network and demonstrates stronger cross-cultural capabilities (Hu, Liu, Zhang & Wang, 2020; Thompson, 2005). Similarly, specific intercultural traits (e.g., cultural flexibility, cultural empathy, social initiative) have been linked to various employee attitudes and perceptions in the workplace, as well as indicators of cross-cultural adjustment (e.g., Peltokorpi, 2008; Ponterotto et al., 2007; Van Der Zee & Van Oudenhoven, 2000; Van Erp, Van der Zee, Giebels, & Van Duijn, 2014; Yakunina, Weigold, Weigold, Hercegovac, & Elsayed, 2012). As individual dispositions may influence workplace integration indirectly through their impact on other factors (e.g., motivation, cultural knowledge; networking), future research that takes into account these dispositions can help us to advance our understanding of what characteristics of international graduates contribute to successful workplace integration and the behaviors that can facilitate more successful integration outcomes.

Further research is also needed exploring how international graduates' experiences of emotion influence their work integration. Several studies have highlighted the integral role that emotion plays in the socialization process (Allen, Eby, Chao & Bauer, 2017; Grant & Nadin, 2007; Wanberg, Choi, & Wanberg, 2012). Hajro et al. (2017) identify eight emotion-focused coping strategies used by skilled immigrants in dealing with workplace challenges (i.e., creating positive social distinctions, preservation, expectation

adjustment, positive reappraisal, resignation, escapism, seeking out social-emotional support, and using alternative resources to strengthen emotional resilience). Given that emotions are intricately linked to self-regulation processes, including one's approach and avoidance tendencies and one's ability to deal effectively with ambiguity and anxiety (Lanaj et al., 2012), research exploring the emotions of international graduates can shed additional light on their integration experience. For example, research examining the emotion-focused coping strategies used during various phases of school to work transitions (e.g., job search, organizational entry, adaptation) may deepen our understanding of international graduates' affective experiences and reactions during the integration process.

Meso-level

As discussed earlier, one of the primary impediments to the successful integration of international graduates into the workforce is the presence of cultural biases and discrimination in organizations' employment processes. The economics literature distinguishes between two types of discrimination: taste-based and statistical (Esses, 2021). While taste-based discrimination is affectively based prejudice against certain groups, statistical discrimination is based on stereotypes regarding a group's performance in a particular area (Esses et al., 1998; Esses, 2021). Although different outcomes of discrimination have been examined in non-immigrant samples (e.g., job dissatisfaction, lower physical health; Jones, Peddie, Gilrane, King & Gray, 2016; Schmitt, Branscombe, Postmes & Garcia, 2014), further study of both the predictors and consequences of discrimination for international graduates is needed. Given their status as immigrants and also often ethnic minorities, international graduates may be particularly vulnerable to acts of discrimination and harassment in the workplace. For example, studies have indicated that during the recent COVID-19 pandemic Asian people in several western countries have experienced stigmatization and "othering" (Coffey, Cook, Farrugia, Threadgold & Burke, 2021; Haokip, 2020; Kwok, 2020, Roberto, Johnson & Rauhaus, 2020). International graduates who are ethnic minorities were also more likely to be viewed as outsiders and to experience bias and harassment in the workplace (Roberto et al., 2020). Further research is needed exploring factors that contribute to this discrimination and harassment faced by international graduates who are ethnic minorities. Post-colonial (Quayson, 2000) and critical race (Ladson-Billings & Tate, 1995) theories may serve as useful conceptual lenses in this work that may help uncover institutional, structural, and cultural factors that contribute to this inequality and enhance our understanding of how policies and power structures can be changed in order to reduce systemic discrimination against international graduates. Likewise, additional research is needed identifying barriers hindering the selection and career development of international graduates in organizations, and what specific programs and policies may be implemented to eliminate these barriers. For example, different methods of applicant screening and assessor training should be examined for their efficacy in reducing discrimination. There is some evidence that interviewer training in assessing job-related qualifications can be used to reduce bias relating to various demographic and socio-cultural differences (e.g., accent, ethnicity, religion, country of birth; Esses et al., 2014; Purkiss, Perrewé, Gillespie, Mayes & Ferris, 2006; Segrest, 2011). Likewise, "name-blind" recruitment (also referred to as "anonymized hiring") also holds promise in potentially reducing discrimination resulting from identifying a candidate's ethnicity or country of origin in screening processes (Banerjee et al., 2018; Government of Canada, 2018); however, this approach requires additional empirical scrutiny.

Although several studies have examined the role of employers, very little research has examined how educational institutions facilitate international students' transition and integration into the workplace. While universities are primarily seen as institutions that enhance knowledge and skill acquisition, they also play a key role in selecting and developing future immigrants (Suter & Jandl, 2008). Moreover, educational institutions and educators can also be instrumental in reducing stereotypes against international students (Tran & Vu, 2016). Thus, more research is needed exploring the role of educational institutions in improving the workforce integration of international graduates. For example, studies should examine what post-secondary institutions can do to collaborate with government agencies and organizations to better tailor programs and workshops to address the needs of international students and improve their work integration and career success (Arthur & Flynn, 2011). Indeed, Cameron et al. (2019) highlight the need for closer employer and university collaboration in order to more proactively support international students' transitions to employment. Likewise, based on interviews and surveys with both international students and employers in Australia, Asia and the United Kingdom, Lawrence (2013) concluded that Australian universities need to collaborate, share market intelligence, and provide more holistic approaches to internships. Educational institutions also need to find ways to improve outreach to international students who, in some cases, may be less aware or willing to avail themselves of available career supports (e.g., Huang & Turner, 2018).

Macro-level

Several studies at the macro-level have emphasized that a country's immigration policies can play a critical role in influencing international graduates' settlement decisions and work integration (Hawthorne, 2010, 2014; Lu et al., 2009; Suter & Jandl, 2008). However, very little research has directly examined the effects of macro-level policies (e.g., immigration policy, government support, labour laws and policies) on international graduates' perceptions and integration experiences. Thus, we do not know how international students' decisions and perceptions are influenced by specific programs/policies, or the extent to which immigration policies combine with other factors across levels (e.g., organizational policies, individual characteristics) to influence the work integration of international graduates. In response to the COVID-19 pandemic, and the impact of travel restrictions on meeting immigration targets, some countries have developed fast-track permanent residence policies that facilitate post-study work visa holders' permanent residence applications (e.g., "New 2021 Resident Visa of New Zealand"; New Zealand Immigration, 2021; "New Pathway to Canadian Permanent Residency", IRCC, 2021). It would be interesting to examine the long-term impact of these policies on different facets of international graduates' work integration (e.g., perceived inclusion, work-related well-being, career mobility).

Most studies examining immigration policies and programs focus on the country level of analysis. However, international students tend to settle unevenly across host countries, and in several countries, specific regions (e.g., states/provinces) have developed their own immigration policies/programs (Lu et al., 2009; Suter & Jandl, 2008). More research on regional or state-level programs and policies should be carried out to identify ways to integrate and retain international graduates in different regions. Related to this point, policy makers and employers may benefit from having a more in-depth understanding how the culture of a host country (and organization) may contribute to international graduates' adjustment and integration. For example, cultural dimensions (e.g., performance orientation, in-group collectivism, power distance) identified in research on the GLOBE (House, Hanges, Javidan, Dorfman & Gupta, 2004) may be used to compare host and home countries' workplace values and practices to identify what cultural differences may present the greatest challenges with respect to international graduates' integration.

Research methodology

Research focusing on international graduates is in its relatively early stages of development; however, this research is somewhat fragmented and it spans across different disciplines and levels of analysis. Because research in this area is multidisciplinary, both qualitative and quantitative methodologies have been applied; however, few studies have attempted to replicate or triangulate findings, or have adopted a mixed-methods approach. Many quantitative studies have used census or national survey data (e.g., Arbeit, & Warren, 2013; Boyd & Tian, 2018). One advantage of using such data is the large sample size and access to objective career data (e.g., earnings); however, using these larger secondary databases may constrain theoretical development and the exploration of international graduates' perceptions of their integration experience. Given the unique characteristics and challenges faced by international graduates, more concerted efforts to develop and test relevant theories and to probe different aspects of their integration experience are needed. Workplace integration has been conceptualized as a dynamic process that follows multiple phases of organizational socialization (Tharenou & Kulik, 2020). In order to better capture the complex temporal nature of this process, more nuanced theory development, and the use of more sophisticated research designs (e.g., longitudinal, mixed-method) are needed that explore the unique and joint effects that specific individual, organizational, and societal factors may exert at different stages of integration.

Conclusion

Although international students represent a critical segment of the global talent pool and boost economic growth in receiving countries, the literature on international graduates' work integration is relatively new and fragmented across disciplines. In this paper, we reviewed and synthesized the literature in this domain by identifying central themes and findings that have emerged at different levels of analysis. Based on our assessment of the literature, we have discussed key limitations and research gaps, and delineate several avenues for future research. We hope that the current review will energize further research in this area, yielding additional evidence-based insights into how to improve the workforce integration of international graduates.

Funding

This work was supported by a CRIW Ignite Research Grant funded by the Centre for Research on Inclusion at Work (CRIW).

Acknowledgments

We thank Nabeel Rustom for his assistance with the literature search and preliminary analysis of the literature. We are also grateful for the helpful comments and suggestions provided by Associate Editor, Jaimee Stuart, and the anonymous reviewers.

References

- Adamuti-Trache, M., & Sweet, R. (2005). Exploring the relationship between educational credentials and the earnings of immigrants. *Canadian Studies in Population*, 32(2), 177–201.
- Adsera, A., & Chiswick, B. R. (2007). Are there gender and country of origin differences in immigrant labor market outcomes across European destinations? *Journal of Population Economics*, 20(3), 495–526.
- Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50(2), 179–211.
- Ajzen, I. (2011). The theory of planned behavior: Reactions and reflections. *Psychology & Health*, 26(9), 1113–1127.
- Akresh, I. R. (2007). US immigrants labor market adjustment: additional human capital investment and earnings growth. *Demography*, 44, 865–881.
- Akresh, I. R. (2008). Occupational trajectories of legal US immigrants: downgrading and recovery. *Population and Development Review*, 34(3), 435–456.
- Allen, T. D., Eby, L. T., Chao, G. T., & Bauer, T. N. (2017). Taking stock of two relational aspects of organizational life: tracing the history and shaping the future of socialization and mentoring research. *Journal of Applied Psychology*, 102, 324–337.
- Altbach, P., & Reisberg, L. (2013). The pursuit of international students in a commercialized world. *International Higher Education*, 73, 2–4.
- Anderson, V., McGrath, T., & Butcher, A. (2014). Asia-born New Zealand-educated business graduates' transition to work. *Asia Pacific Journal of Education*, 34(1), 65–79.
- Anisef, P., Sweet, R., & Frempong, G. (2003). Labour market outcomes of immigrant and racial minority university graduates in Canada. *Journal of International Migration and Integration*, 4(4), 499–522.
- Arbeit, C. A., & Warren, J. R. (2013). Labor market penalties for foreign degrees among college educated immigrants. *Social Science Research*, 42(3), 852–871.
- Arkoudis, S., Hawthorne, L., Baik, C., Hawthorne, G., O'Loughlin, K., Leach, D., & Bexley, E. (2009). *The impact of English language proficiency and workplace readiness on employment outcomes and performance of tertiary international students*. 45. Melbourne: Centre for the Study of Higher Education, University of Melbourne.
- Artess, J., Hooley, T., & Mellors-bourne, R. (2017). *Employability: A Review of the Literature 2012 to 2016*. Heslington, UK: Higher Education Academy (HEA).
- Arthur, N., & Flynn, S. (2011). Career development influences of international students who pursue permanent immigration to Canada. *International Journal for Educational and Vocational Guidance*, 11(3), 221–237.

- Arthur, N., & Flynn, S. (2013). International students views of transition to employment and immigration. *Canadian Journal of Career Development*, 12(1), 28–37.
- Arthur, N. (2007). Career planning and decision-making needs of international students. In H. Singaravelu, & M. Pope (Eds.), *A handbook for counselling international students in the United States* (pp. 37–56). Alexandria, VA: American Counseling Association.
- Arthur, N. (2008). Counseling international students. In P. Pedersen, J. G. Draguns, W. J. Lonner, & J. E. Trimble (Eds.), *Counseling across cultures* (6th ed., pp. 275–290). Thousand Oaks, CA: Sage.
- Baas, M. (2019). The Education-migration industry: International students, migration policy and the question of skills. *International Migration*, 57(3), 222–234.
- Balibar, E. (1992). Is there a “neo-racism”? In W. Balibar (Ed.), *Race, nation, class: Ambiguous identities* (pp. 17–28). New York: Verso.
- Bandura, A. (2001). Social cognitive theory: an agentic perspective. *Annual Review of Psychology*, 52(1), 1–26.
- Banerjee, R., Reitz, J. G., & Oreopoulos, P. (2018). Do large employers treat racial minorities more fairly? An analysis of Canadian field experiment data. *Canadian Public Policy*, 44(1), 1–12.
- Becker, G. S. (1962). Investment in human capital: a theoretical analysis. *Journal of Political Economy*, 70, 9–49.
- Becker, G. S. (1964). *Human Capital: A Theoretical and Empirical Analysis with Special Reference to Education*. New York: National Bureau of Economic Research.
- Becker, G. S. (1993). Nobel lecture: the economic way of looking at behavior. *Journal of Political Economy*, 101, 385–409.
- Becker, G. S. (2009). *Human capital: A theoretical and empirical analysis, with special reference to education*. Chicago, IL: University of Chicago Press.
- Biddle, B. J. (1986). Recent development in role theory. *Annual Review of Sociology*, 12, 67–92.
- Blackmore, J., Gribble, C., Farrell, L., Rahimi, M., Arber, R., & Devlin, M. (2014). *Australian international graduates and the transition to employment*. Deakin University.
- Blackmore, J., Gribble, C., & Rahimi, M. (2017). International education, the formation of capital and graduate employment: Chinese accounting graduates’ experiences of the Australian labour market. *Critical Studies in Education*, 58(1), 69–88.
- Blackmore, J., & Rahimi, M. (2019). How ‘best fit’ excludes international graduates from employment in Australia: A Bourdieusian perspective. *Journal of Education and Work*, 32(5), 436–448.
- Booth, A., Leigh, A., & Varganova, E. (2010). Does racial and ethnic discrimination vary across minority groups? Evidence from a field experiment (IZA Discussion Paper No. 4947).
- Bourdieu, P. (1977). *Outline of a theory of practice*, 16. New York, NY: Cambridge University Press.
- Bourdieu, P. (1990). *Other Words: Essays Towards a Reflexive Sociology*. Menlo Park, CA: Stanford University Press.
- Bourdieu, P. (1986). The forms of capital. In J. Richardson (Ed.), *Handbook of theory and research for the sociology of education* (pp. 241–258). New York: Greenwood Press.
- Boyd, M., & Tian, S. (2018). Is STEM education portable? Country of education and the economic integration of STEM immigrants. *Journal of International Migration and Integration*, 19(4), 965–1003.
- Bozionelos, N., Bozionelos, G., Kostopoulos, K., Shyong, C. H., Baruch, Y., & Zhou, W. (2015). International graduate students’ perceptions and interest in international careers. *International Journal of Human Resource Management*, 26(11), 1428–1451.
- Bratman, M. E. (1999). *Faces of intention: selected essays on intention and agency*. New York, NY: Cambridge University Press.
- Brown, P. (2000). The Globalisation of Positional Competition? *Sociology*, 34(4), 633–653.
- Brown, P., Hesketh, A., & Williams, S. (2004). *The mismanagement of talent: employability and jobs in the knowledge economy*. Oxford, UK: Oxford University Press on Demand.
- Bruck-Lee, V., Khoury, H. A., Nixon, A. E., Goh, A., & Spector, P. E. (2009). Replicating and extending past personality/job satisfaction meta-analyses. *Human Performance*, 22(2), 156–189.
- Buddelmeyer, H., Van de Ven, J., & Zakirova, R. (2013). *Graduate employment outcomes for international students*. Melbourne Institute of Applied Economic and Social Research, University of Melbourne.
- Cameron, R., Farivar, F., & Coffey, J. (2019). International graduates host country employment intentions and outcomes: Evidence from two Australian universities. *Journal of Higher Education Policy and Management*, 41(5), 550–568.
- Carlson, H. K., & McHenry, M. A. (2006). Effect of accent and dialect on employability. *Journal of Employment Counseling*, 43(2), 70–83.
- Carr, S. C., Inkson, K., & Thorn, K. (2005). From global careers to talent flow: Reinterpreting ‘brain drain’. *Journal of World Business*, 40(4), 386–398.
- Chen, Z. J., & Skuterud, M. (2018). Relative labour market performance of former international students: Evidence from the National Graduates Survey. *Canadian Public Policy*, 44(3), 207–225.
- Cheung, S. Y., & Phillimore, J. (2017). Gender and refugee integration: a quantitative analysis of integration and social policy outcomes. *Journal of Social Policy*, 46(2), 211–230.
- Chiaburu, D. S., Oh, I.-S., Berry, C. M., Li, N., & Gardner, R. G. (2011). The five-factor model of personality traits and organizational citizenship behaviors: a meta-analysis. *Journal of Applied Psychology*, 96(6), 1140–1166.
- Chiou, B. (2017). Two-step migration: A comparison of Australia’s and New Zealand’s policy development between 1998 and 2010. *Asian and Pacific Migration Journal*, 26(1), 84–107.
- Chira, S., Barber, P., & Belkhdja, C. (2013). Dreaming big, coming up short: The challenging realities of international students and graduates in Atlantic Canada. The Atlantic Metropolis Centre’s Working Papers Series, Working Paper No.47–2013.
- Chiswick, B. R., & Miller, P. W. (2009). The international transferability of immigrants’ human capital. *Economics of Education Review*, 28(2), 162–169.
- Choi, Y., Crossman, E., & Hou, F. (2021). International students as a source of labour supply: transition to permanent residency. *Economic and Social Reports. Statistics Canada*.
- Choi, Y., Hou, F., & Chan, P. (2021). Early earnings trajectories of international students after graduation from postsecondary programs. *Economic and Social Reports. Statistics Canada*.
- Coffey, J., Cook, J., Farrugia, D., Threadgold, S., & Burke, P. J. (2021). Intersecting marginalities: International students’ struggles for “survival” in COVID-19. *Gender, Work & Organization*, 28(4), 1337–1351.
- Coffey, J., Farivar, F., & Cameron, R. (2018). The job seeking experiences of international graduates in the host country: Australia’s lost opportunity? *International Journal Human Resource Management*.
- Coleman, S. J. (1988). Social capital in the creation of human capital. *American Journal of Sociology*, 94(Suppl), S95–S120.
- Collins, R. (1971). Functional and conflict theories of educational stratification. *American Sociological Review*, 36, 1002–1019.
- Conrad, H., & Meyer-Ohle, H. (2019a). Overcoming the ethnocentric firm?—foreign fresh university graduate employment in Japan as a new international human resource development method. *The International Journal of Human Resource Management*, 30(17), 2525–2543.
- Conrad, H., & Meyer-Ohle, H. (2019b). Transnationalization of a recruitment regime: skilled migration to Japan. *International Migration*, 57(3), 250–265.
- Considine, M. (1994). *Public Policy: A Critical Approach*. South Melbourne: Macmillan.
- Considine, M. (2005). *Making Public Policy*. Cambridge: Polity Press.
- Constant, A. F., Nottmeyer, O., & Zimmermann, K. F. (2009). Cultural integration in Germany. In Y. Algan, A. Bisin, A. Manning, & T. Verdier (Eds.), *Cultural Integration of Immigrants in Europe* (pp. 69–124). UK: Oxford University Press.
- Cranmer, S. (2006). Enhancing graduate employability: best intentions and mixed outcomes. *Studies in Higher Education*, 31(2), 169–184.
- Crossman, E., Hou, F., & Picot, G. (2020). Two-step immigration selection: Recent trends in immigrant labour market outcomes. *Economic Insights*. Statistics Canada. Catalogue no. 11-626-X -2020009-No. 111.
- Dam, H., Chan, J., & Wayland, S. (2018). Missed opportunity: International students in Canada face barriers to permanent residence. *Journal of International Migration and Integration*, 19(4), 891–903.
- Dancygier, R. M., & Laitin, D. D. (2014). Immigration into Europe: economic discrimination, violence, and public policy. *Annual Review of Political Science*, 17, 43–64.
- DeFillippi, R. J., & Arthur, M. B. (1994). The boundaryless career: a competency-based perspective. *Journal of Organizational Behavior*, 15(4), 307–324.
- Doherty, N. (2013). Understanding the self-initiated expatriate: A review and directions for future research. *International Journal of Management Reviews*, 15(4), 447–469.

- Dowling, E., & Harvie, D. (2014). Harnessing the social: state, crisis and (big) society. *Sociology*, 48(5), 869–886.
- Dulebohn, J. H., Bommer, W. H., Liden, R. C., Brouer, R. L., & Ferris, G. R. (2012). A meta-analysis of antecedents and consequences of leader-member exchange: integrating the past with an eye toward the future. *Journal of Management*, 38(6), 1715–1759.
- Educationdata (2020). International Student Enrollment Statistics. Retrieved from (<https://educationdata.org/international-student-enrollment-statistics>).
- Van Erp, K. J., Van der Zee, K. I., Giebels, E., & Van Duijn, M. A. (2014). Lean on me: The importance of one's own and partner's intercultural personality for expatriate's and expatriate spouse's successful adjustment abroad. *European Journal of Work and Organizational Psychology*, 23(5), 706–728.
- Esses, V., Sutter, A., Ortiz, A., Luo, N., Cui, J., & Deacon, L. (2018). Retaining international students in Canada post-graduation: Understanding the motivations and drivers of the decision to stay. *CBIE Research in Brief*, 8.
- Esses, V. M. (2021). Prejudice and discrimination toward immigrants. *Annual Review of Psychology*, 72, 503–531.
- Esses, V. M., Bennett-AbuAyyash, C., & Lapshina, N. (2014). How discrimination against ethnic and religious minorities contributes to the underutilization of immigrants' skills. *Policy Insights from the Behavioral and Brain Sciences*, 1(1), 55–62.
- Esses, V. M., Jackson, L. M., & Armstrong, T. L. (1998). Intergroup competition and attitudes toward immigrants and immigration: An instrumental model of group conflict. *Journal of Social Issues*, 54(4), 699–724.
- Farivar, F., Coffey, J., & Cameron, R. (2019). International graduates and the change of initial career mobility intentions. *Personnel Review*, 48(4), 1061–1078.
- Favell, A., Feldblum, M., & Smith, P. (2006). Mobility, migration, and technology workers: an introduction. *Knowledge, Technology and Policy*, 19, 3–6.
- Ferdman, B. M. (2017). Paradoxes of inclusion: understanding and managing the tensions of diversity and multiculturalism. *The Journal of Applied Behavioral Science*, 53(2), 235–263.
- Firang, D. (2020). The impact of COVID-19 pandemic on international students in Canada. *International Social Work*, 63(6), 820–824.
- Fogg, P., & Harrington, E. (2012). *The earnings of foreign-educated college graduates: an examination of the determinants of the hourly earnings of college-educated immigrants*. Drexel University.
- Forde, C., & MacKenzie, R. (2010). The ethical agendas of employment agencies towards migrant workers in the UK: Deciphering the codes. *Journal of Business Ethics*, 97, 31–41.
- Fossland, T. (2013). Negotiating future careers: a relational perspective on skilled migrants' labour market participation. *Journal of Management Development*, 67(5), 276–283.
- Franco, M., Hsiao, Y. S., Gnlika, P. B., & Ashby, J. S. (2019). Acculturative stress, social support, and career outcome expectations among international students. *International Journal for Educational and Vocational Guidance*, 19(2), 275–291.
- Frank, K., & Hou, F. (2018). Over-education and well-being: how does education-occupation mismatch affect the life satisfaction of university-educated immigrant and non-immigrant workers? *Ethnicity & Health*, 23(8), 884–901.
- Frazis, H. (2002). Human capital, signaling, and the pattern of returns to education. *Oxford Economic Papers*, 54, 298–320.
- Fritz, M. V., Chin, D., & DeMarinis, V. (2008). Stressors, anxiety, acculturation, and adjustment among international and North American students. *International Journal of Intercultural Relations*, 32, 244–259.
- Fuller, B. J., & Marler, L. E. (2009). Change driven by nature: a meta-analytic review of the proactive personality literature. *Journal of Vocational Behavior*, 75(3), 329–345.
- Geddie, K. (2013). The transnational ties that bind: relationship considerations for graduating international science and engineering research students. *Population, Space and Place*, 19(2), 196–208.
- De Genova, N. (2013). "We are of the connections": migration, methodological nationalism, and "militant research". *Postcolonial Studies*, 16(3), 250–258.
- Government of Canada (2018). Anonymized recruitment pilot project final report. Retrieved from (https://www.canada.ca/en/public-service-commission/services/publications/Name-blind-recruitment-pilot-project.html#toc_e).
- Grant, P. R., & Nadin, S. (2007). The credentialing problems of foreign trained personnel from Asia and Africa intending to make their home in Canada: a social psychological perspective. *Journal of International Migration and Integration*, 8, 141–162.
- Gribble, C. (2014). Employment, work placements and work integrated learning of international students in Australia. *International Education Association of Australia, Research Digest*, 2.
- Gribble, C., Rahimi, M., & Blackmore, J. (2017). International students and post-study employment: the impact of university and host community engagement on the employment outcomes of international students in Australia. *International Student Connectedness and Identity* (pp. 15–39). Singapore: Springer.
- Gümüş, S., Gök, E., & Esen, M. (2020). A review of research on international student mobility: science mapping the existing knowledge base. *Journal of Studies in International Education*, 24(5), 495–517.
- Hajro, A., Stahl, G. K., Clegg, C. C., & Lazarova, M. B. (2019). Acculturation, coping, and integration success of international skilled migrants: an integrative review and multilevel framework. *Human Resource Management Journal*, 29(3), 328–352.
- Hajro, A., Zilinskaite, M., & Stahl, G. (2017). Acculturation of highly-qualified migrants at the workplace: The importance of individual coping strategies and organizational climate for inclusion. Academy of Management Meeting, Atlanta, USA, 4–8 August, 2017, 13666.
- Hall, P. A., & Soskice, D. (2001). An introduction to varieties of capitalism. In P. A. Hall, & D. Soskice (Eds.), *Varieties of capitalism* (pp. 1–68). Oxford: Oxford University Press.
- Haokip, T. (2020). From 'Chinky' to 'coronavirus': Racism against Northeast Indians during the Covid-19 pandemic. *Asian Ethnicity*, 1–10.
- Hari, A., Nardon, L., & Zhang, H. (2021). A transnational lens into international student experiences of the COVID-19 pandemic. *Global Networks*. <https://doi.org/10.1111/glob.12332>
- Harrison, G., & Ip, R. (2013). Extending the terrain of inclusive education in the classroom to the field: International students on placement. *Social Work Education*, 32(2), 230–243.
- Hartog, J. (2000). Over-education and earnings: where are we, where should we go? *Economics of Education Review*, 19(2), 131–147.
- Harvey, D. (2017). *Marx, Capital and the Madness of Economic Reason*. London: Profile Books.
- Hawthorne, L. (2010). How valuable is "two-step migration"? Labor market outcomes for international student migrants to Australia. *Asian and Pacific Migration Journal*, 19(1), 5–36.
- Hawthorne, L. (2014). Indian students and the evolution of the study-migration pathway in Australia. *International Migration*, 52(2), 3–19.
- Hawthorne, L. (2018). Attracting and retaining international students as skilled migrants. In M. Czaika (Ed.), *High-skilled migration: Drivers and policies* (pp. 195–220). UK: Oxford University Press.
- Hervik, P. (2004). Anthropological perspectives on the new racism in Europe. *Ethnos*, 69(2), 149–155.
- Hou, F., & Bonikowska, A. (2018). Selections before the selection: Earnings advantages of immigrants who were former skilled temporary foreign workers in Canada. *International Migration Review*, 52(3), 695–723.
- Hou, F., & Lu, Y. (2017). International students, immigration and earnings growth: the effect of a pre-immigration host-country university education. *IZA Journal of Development and Migration*, 7(5), 1–24.
- House, R. J., Hanges, P. J., Javidan, M., Dorfman, P. W., & Gupta, V. (Eds.). (2004). *Culture, leadership, and organizations: The GLOBE study of 62 societies*. Thousand Oaks, Calif: Sage Publications.
- Hu, S., Liu, H., Zhang, S., & Wang, G. (2020). Proactive personality and cross-cultural adjustment: roles of social media usage and cultural intelligence. *International Journal of Intercultural Relations*, 74, 42–57.
- Huang, I. Y., Raimo, V., & Humfrey, C. (2016). Power and control: managing agents for international student recruitment in higher education. *Studies in Higher Education*, 41(8), 1333–1354.
- Huang, R., & Turner, R. (2018). International experience, universities support and graduate employability—perceptions of Chinese international students studying in UK universities. *Journal of Education and Work*, 31(2), 175–189.
- IIE (2020). Project Atlas: Global Mobility Trends. Retrieved from (<https://www.iie.org/en/Research-and-Insights/Project-Atlas/Explore-Data>).

- IRCC (2017). The employer's roadmap to hiring and retaining internationally trained workers. Retrieved from (<https://www.canada.ca/content/dam/ircc/migration/ircc/english/pdf/pub/employer-roadmap-en.pdf>).
- IRCC (2021). New pathway to permanent residency for over 90,000 essential temporary workers and international graduates. Accessed on October 24, 2021 from (<https://www.canada.ca/en/immigration-refugees-citizenship/news/2021/04/new-pathway-to-permanent-residency-for-over-90000-essential-temporary-workers-and-international-graduates.html>).
- Jenei, K., Cassidy-Matthews, C., Virk, P., Lulie, B., & Closson, K. (2020). Challenges and opportunities for graduate students in public health during the COVID-19 pandemic. *Canadian Journal of Public Health*, 111(3), 408–409.
- Jiang, X., & Kim, D. (2019). The price of being international: career outcomes of international Master's recipients in US labor market. *Journal of International Students*, 9(3), 732–757.
- Jiang, Z. (2017). Proactive personality and career adaptability: the role of thriving at work. *Journal of Vocational Behavior*, 98, 85–97.
- Jones, K. P., Peddie, C. I., Gilrane, V. L., King, E. B., & Gray, A. L. (2016). Not so subtle: a meta-analytic investigation of the correlates of subtle and overt discrimination. *Journal of Management*, 42(6), 1588–1613.
- Jordan, J. V., Kaplan, A. G., Miller, J. B., Stiver, I. P., & Surrey, J. L. (1991). *Women's growth in connection*. New York, NY: Guilford.
- Joseph, M. G. C. (2016). From students to permanent residents: the role of social networks in the immigration transition of Indians in New Zealand. *Asian and Pacific Migration Journal*, 25(2), 168–190.
- Judge, T. A., Heller, D., & Mount, M. K. (2002). Five-factor model of personality and job satisfaction: a meta-analysis. *Journal of Applied Psychology*, 87(3), 530–541.
- Khan, M. (2019). Contested Ground: network Governance in Australia's Migration Industry. *International Migration*, 57(3), 295–309.
- Kim, J. (2016). Global cultural capital and global positional competition: international graduate students' transnational occupational trajectories. *British Journal of Sociology of Education*, 37(1), 30–50.
- Kim, S. (2015). The influence of social relationships on international students' intentions to remain abroad: multi-group analysis by marital status. *The International Journal of Human Resource Management*, 26(14), 1848–1864.
- Kim, S., & Han, S. J. (2017). Global talent flow: International graduate students' career decisions after graduation. *Talent Development and the Global Economy: Perspectives from Special Interest Groups*, 7–27.
- Kwok, H. (2020). Beyond the anti-racist reason: A postcolonial perspective on pandemic politics. *Health Sociology Review*, 29(2), 122–130.
- Ladson-Billings, G., & Tate, W. (1995). Toward a critical race theory of education. *Teachers College Record*, 97(1), 47–68.
- Lanaj, K., Chang, C.-H. "D.", & Johnson, R. E. (2012). Regulatory focus and work-related outcomes: A review and meta-analysis. *Psychological Bulletin*, 138(5), 998–1034.
- Lashari, S. A., Kaur, A., & Awang-Hashim, R. (2018). Home away from home-the role of social support for international students' adjustment. *Malaysian Journal of Learning and Instruction*, 15(2), 33–54.
- Lawrence, R. (2013). *AUIDF Improving the employment outcomes of international students*. Melbourne: Prospect Research and Marketing: Australian Universities International Director's Forum (AUIDF).
- Lee, E. S., Szkudlarek, B., Nguyen, D. C., & Nardon, L. (2020). Unveiling the Canvas Ceiling: a multidisciplinary literature review of refugee employment and workforce integration. *International Journal of Management Reviews*, 22(2), 193–216.
- Lent, R. W., Brown, S. D., & Hackett, G. (1994). Toward a unifying social cognitive theory of career and academic interest, choice and performance. *Journal of Vocational Behavior*, 45, 79–122.
- Lent, R. W., Brown, S. D., & Hackett, G. (2000). Contextual supports and barriers to career choice: a social cognitive analysis. *Journal of Counseling Psychology*, 47, 36–49.
- Lent, R. W., Brown, S. D., & Hackett, G. (1996). Career development from a social cognitive perspective. In D. Brown, & L. Brooks (Eds.), *Career Choice and Development* (3rd ed., pp. 373–421). San Francisco, CA: Jossey-Bass.
- Lent, R. W., Brown, S. D., Sheu, H.-B., Schmidt, J., Brenner, B. R., Gloster, C. S., ... Treisman, D. (2005). Social cognitive predictors of academic interests and goals in engineering: utility for women and students at historically black universities. *Journal of Counseling Psychology*, 52(1), 84–92.
- Lien, D. (2005). International accreditation and brain drain: a simple model. *Economics of Education Review*, 25, 335–340.
- Lu, Y., & Hou, F. (2015). International students who become permanent residents in Canada. *Insights on Canadian Society*. Statistics Canada Catalogue (75-006).
- Lu, Y., & Hou, F. (2019). Over-Education among University-Educated Immigrants in Canada and the United States. *Analytical Studies Branch Research Paper Series*. Statistics Canada.
- Lu, Y., Zong, L., & Schissel, B. (2009). To stay or return: migration intentions of students from People's Republic of China in Saskatchewan, Canada. *Journal of International Migration and Integration*, 10(3), 283–310.
- Madge, C., Raghuram, P., & Noxolo, P. (2009). Engaged pedagogy and responsibility: a postcolonial analysis of international students. *Geoforum*, 40, 34–45.
- Maury, O. (2017). Student-migrant-workers: temporal aspects of precarious work and life in Finland. *Nordic Journal of Migration Research*, 7(4), 224–232.
- Maury, O. (2020). Between a promise and a salary: student-migrant-workers' experiences of precarious labour markets. *Work, Employment and Society*, 34(5), 809–825.
- Miller, J. B., & Stiver, I. P. (1997). *The healing connection: how women form relationships in therapy and in life*. Boston: Beacon Press.
- Mok, K. H., & Chan, S. J. (2020). Promoting global cities status: talent attraction and retention in. *Asia. Globalisation, Societies and Education*, 18(1), 1–6.
- Morris-Lange, S. (2019). Retention of international students/graduates in Canada, Germany, and the Netherlands. ICMPP Working Paper. Retrieved from: (<https://www.icmppd.org/file/download/48186/file/Labour0Migration0Working0Paper0Retention0of0International0Students0EN.pdf>).
- Moskal, M. (2017). International students pathways between open and closed borders: towards a multi-scalar approach to educational mobility and labour market openness. *International Migration*, 55(3), 126–138.
- Mosneaga, A., & Winther, L. (2013). Emerging talents? International students before and after their career start in Denmark. *Population, Space and Place*, 19(2), 181–195.
- Musumba, M., Jin, Y. H., & Mjelde, J. W. (2011). Factors influencing career location preferences of international graduate students in the United States. *Education Economics*, 19(5), 501–517.
- New Zealand Immigration (2021). New 2021 Resident Visa of New Zealand. Retrieved from (<https://www.immigration.govt.nz/new-zealand-visas/already-have-a-visa/one-off-residence-visa>).
- Nourpanah, S. (2019). Drive-by education: the role of vocational courses in the migration projects of foreign nurses in Canada. *Journal of International Migration and Integration*, 20(4), 995–1011.
- Nunes, S., & Arthur, N. (2013). International students' experiences of integrating into the workforce. *Journal of Employment Counseling*, 50(1), 34–45.
- OECD. (2001). The well-being of Nations: The role of human and social capital. Paris.
- OECD. (2020). *Education at a Glance 2020: OECD Indicators*. Paris: OECD Publishing. <https://doi.org/10.1787/69096873-en>
- OECD/European Union. (2019). Immigrant skills and labour market integration. *Settling In 2018: Indicators of Immigrant Integration*. Paris/European Union, Brussels: OECD Publishing.
- Oreopoulos, P. (2011). Why do skilled immigrants struggle in the labor market? A field experiment with thirteen thousand resumes. *American Economic Journal: Economic Policy*, 3(4), 148–171.
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Systematic Reviews*, 10(89). <https://doi.org/10.1186/s13643-021-01626-4>
- Papademetriou, D. G., Sommerville, W., & Tanaka, H. (2009). Talent in the 21st century. In Bertelsmann Stiftung, & Migration Policy Institute (Eds.), *Talent, Competitiveness and Migration: The Transatlantic Council on Migration* (pp. 215–266). Gütersloh, Germany: Verlag Bertelsmann Stiftung.
- Paulsen, M. B. (2001). The economics of human capital and investment in higher education. In M. B. Paulsen, & J. C. Smart (Eds.), *The finance of higher education: Theory, research, policy, and practice* (pp. 55–94). New York, NY: Agathon Press.
- Pegg, A., Waldoock, J., Hendy-Isaac, S., & Lawton, R. (2012). *Pedagogy for employability*. York: The Higher Education Academy.

- Peltokorpi, V. (2008). Cross-cultural adjustment of expatriates in Japan. *The International Journal of Human Resource Management*, 19(9), 1588–1606.
- Peng, Y. (2020). From migrant student to migrant employee: three models of the school-to-work transition of mainland Chinese in Hong Kong. *Population, Space and Place*, 26(4), Article e2283.
- Phelps, E. S. (1972). The statistical theory of racism and sexism. *The American Economic Review*, 62(4), 659–661.
- Picot, G., & Hou, F. (2019). Skill utilization and earnings of STEM-educated immigrants in Canada: Differences by degree level and field of study. *Analytical Studies Branch Research Paper Series Statistics Canada*.
- Ponterotto, J. G., Costa-Wofford, C. I., Brobst, K. E., Spelliscy, D., Kacanski, J. M., Scheinholtz, J., & Martinez, D. (2007). Multicultural personality dispositions and psychological well-being. *The Journal of Social Psychology*, 147(2), 119–135.
- Popadiuk, N. E., & Arthur, N. M. (2014). Key relationships for international student university-to-work transitions. *Journal of Career Development*, 41(2), 122–140.
- Portes, A. (1998). Social capital: Its origins and applications in modern sociology. *Annual Review of Sociology*, 24, 1–24.
- Purkiss, S. L. S., Perrewé, P. L., Gillespie, T. L., Mayes, B. T., & Ferris, G. R. (2006). Implicit sources of bias in employment interview judgments and decisions. *Organizational Behavior and Human Decision Processes*, 101(2), 152–167.
- Quayson, A. (2000). *Post-colonialism: theory, practice or process*. Cambridge: Polity.
- Ra, Y. A., & Trusty, J. (2017). Impact of social support and coping on acculturation and acculturative stress of East Asian international students. *Journal of Multicultural Counseling and Development*, 45(4), 276–291.
- Rajani, N., Ng, E. S., & Groutsis, D. (2018). From India to Canada: an autoethnographic account of an International Student's Decision to Settle as a Self-Initiated Expatriate. *Canadian Ethnic Studies*, 50(1), 129–148.
- Roberto, K. J., Johnson, A. F., & Rauhaus, B. M. (2020). Stigmatization and prejudice during the COVID-19 pandemic. *Administrative Theory and Praxis*, 42(3), 364–378.
- Robertson, S. (2011). Student switchers and the regulation of residency: the interface of the individual and Australia's immigration regime. *Population, Space and Place*, 17(1), 103–115.
- Robertson, S., & Runganaikaloo, A. (2014). Lives in limbo: Migration experiences in Australia's education–migration nexus. *Ethnicities*, 14(2), 208–226.
- Rubb, S. (2003). Overeducation in the labor market: a comment and re-analysis of a meta-analysis. *Economics of Education Review*, 22(6), 621–629.
- Ruhs, M. (2008). Economic research and labour immigration policy. *Oxford Review of Economic Policy*, 24(3), 404–427.
- Ruiz, N.G. & Budiman, A. (2018). Number of foreign college students staying and working in U.S. after graduation surges. Accessed on October 30, 2021 from (<https://www.pewresearch.org/global/2018/05/10/number-of-foreign-college-students-staying-and-working-in-u-s-after-graduation-surges/>).
- Salgado, J. F. (2002). The Big Five personality dimensions and counterproductive behaviors. *International Journal of Selection and Assessment*, 10(1–2), 117–125.
- Sanganjanavanich, V. F., Lenz, A. S., & Cavazos, J., Jr. (2011). International students' employment search in the United States: a phenomenological study. *Journal of Employment Counseling*, 48(1), 17–26.
- Schmitt, M. T., Branscombe, N. R., Postmes, T., & Garcia, A. (2014). The consequences of perceived discrimination for psychological well-being: a meta-analytic review. *Psychological Bulletin*, 140(4), 921–948.
- Schultz, T. (1961). Investment in human capital. *American Economic Review*, 51(1), 1–17.
- Segrest, S. L. (2011). Rater error bias training in the employment interview and racioethnicity biased perceptions. *Journal of Academic and Business Ethics*, 3, 1–18.
- Shchegolev, V., Lassleben, H., & Martinez, A. (2016). Employment opportunities and barriers on the Russian and German labor market for international graduates with domestic degrees. *Business Perspectives and Research*, 4(2), 93–110.
- She, Q., & Wotherspoon, T. (2013). International student mobility and highly skilled migration: A comparative study of Canada, the United States, and the United Kingdom. *SpringerPlus*, 2(1), 1–14.
- Shen, Y. J., & Herr, E. L. (2004). Career placement concerns of international graduate students: A qualitative study. *Journal of Career Development*, 31(1), 15–29.
- Shirmohammadi, M., Beigi, M., & Stewart, J. (2019). Understanding skilled migrants' employment in the host country: a multidisciplinary review and a conceptual model. *The International Journal of Human Resource Management*, 30(1), 96–121.
- Shore, L. M., Cleveland, J. N., & Sanchez, D. (2018). Inclusive workplaces: a review and model. *Human Resource Management Review*, 28(2), 176–189.
- Shore, L. M., Randel, A. E., Chung, B. G., Dean, M. A., Ehrhart, K. H., & Singh, G. (2011). Inclusion and diversity in work groups: A review and model for future research. *Journal of Management*, 37, 1262–1289.
- Shumilova, Y., & Cai, Y. (2016). Three approaches to competing for global talent: Role of higher education. In K. Bista, & C. Foster (Eds.), *Global Perspectives and Local Challenges Surrounding International Student Mobility* (pp. 114–135). Hershey, PA, USA: IGI Global.
- Silvey, R., & Lawson, V. (1999). Placing the migrant. *Annals of the Association of American Geographers*, 81, 121–132.
- Sorrenti, G. (2017). The Spanish or the German apartment? Study abroad and the acquisition of permanent skills. *Economics of Education Review*, 60, 142–158.
- Statista (2021a). Number of study permit holders with a valid permit in Canada from 2000 to 2020. Retrieved from (<https://www.statista.com/statistics/555117/number-of-international-students-at-years-end-canada-2000-2014/>).
- Statista (2021b). Education in Canada - Statistics & Facts. Retrieved from (<https://www.statista.com/topics/2863/education-in-canada/>).
- Statistics Canada (2020). International students accounted for all of the growth in postsecondary enrolments in 2018/2019. Retrieved from (<https://www150.statcan.gc.ca/n1/daily-quotidien/201125/dq201125e-eng.htm>).
- Super, D. E. (1969). Vocational development theory: Persons, positions, and processes. *Counseling Psychologist*, 1, 2–9.
- Suter, B., & Jandl, M. (2008). Train and retain: National and regional policies to promote the settlement of foreign graduates in knowledge economies. *Journal of International Migration and Integration*, 9(4), 401–418.
- Sweetman, A., & McBride, S. (2004). Postsecondary field of study and the Canadian labour market outcomes of immigrants and non-immigrants. *Analytical Studies Branch research paper series*. Statistics Canada, Catalogue no. 11F0019MIE – No. 233.
- Sørensen, N. N., & Gammeltoft-Hansen, T. (2012). *The Migration Industry and Future Directions for Migration Policy*. Danish Institute for International Studies.
- Tazzioli, M. (2015). Troubling mobilities: Foucault and the hold over 'unruly' movements and life-time. In M. Tazzioli, S. Fuggle, & Y. Lanci (Eds.), *Foucault and the History of Our Present* (pp. 159–175). Palgrave Macmillan.
- Tharenou, P. (2015). Chinese international business graduates: a career dilemma: Repatriate or stay? *Journal of Management and Organization*, 21(1), 37–59.
- Tharenou, P., & Kulik, C. T. (2020). Skilled Migrants Employed in Developed, Mature Economies: From Newcomers to Organizational Insiders. *Journal of Management*, 46(6), 1156–1181.
- Thompson, J. A. (2005). Proactive Personality and Job Performance: A Social Capital Perspective. *Journal of Applied Psychology*, 90(5), 1011–1017.
- Toren, N. (1976). Return to Zion. *Social Forces*, 54(3), 546–558.
- Tran, L. T., Rahimi, M., Tan, G., Dang, X. T., & Le, N. (2020). Post-study work for international graduates in Australia: opportunity to enhance employability, get a return on investment or secure migration? *Globalisation, Societies and Education*, 18(5), 495–510.
- Tran, L. T., & Vu, T. P. (2016). 'I'm not like that, why treat me the same way?' The impact of stereotyping international students on their learning, employability and connectedness with the workplace. *The Australian Educational Researcher*, 43(2), 203–220.
- UNESCO (2019). Outbound internationally mobile students by host region. Accessed on November 10, 2019 from (<http://data.uis.unesco.org/Index.aspx?queryid=172>).
- Veit, S., & Thijsen, L. (2019). Almost identical but still treated differently: hiring discrimination against foreign-born and domestic-born minorities. *Journal of Ethnic and Migration Studies*, 1–20.
- Walker, M. (2008). A human capabilities framework for evaluating student learning. *Teaching in Higher Education*, 13, 477–487.
- Wall, T., Tran, L. T., & Soejatminah, S. (2018). Inequalities and agencies in workplace learning experiences: International student perspectives. *Vocations and Learning*, 10(2), 141–156.
- Wanberg, C. R., & Choi, Y. (2012). Moving forward: Next steps for advancing the research and practice of employee socialization. In C. R. Wanberg (Ed.), *The Oxford handbook of organizational socialization* (pp. 339–346). New York, NY: Oxford University Press.

- Wayland, S. & Sheffield, D. (2014). Government as employer of skilled immigrants. Retrieved from (http://www.hireimmigrants.ca/wp-content/uploads/Government_as_Employer.pdf).
- Wilkinson, L. (2008). Labor Market Transitions of Immigrant-Born, Refugee-Born, and Canadian-Born Youth. *Canadian Review of Sociology*, 45(2), 151–176.
- Wilton, N. (2012). The impact of work placements on skills development and career outcomes for business and management graduates. *Studies in Higher Education*, 37(5), 603–620.
- Yakunina, E. S., Weigold, I. K., Weigold, A., Hercegovac, S., & Elsayed, N. (2012). The multicultural personality: Does it predict international students' openness to diversity and adjustment? *International Journal of Intercultural Relations*, 36(4), 533–540.
- Yang, P. (2017). Flexible citizens or disconnected transmigrants? Chinese student-turned-migrants in Singapore and their discourse on mobility, flexibility, and identity. In L. T. Tran, & C. Gomes (Eds.), *International student connectedness and identity* (pp. 227–242). Singapore: Springer.
- Yeoh, B. S., Willis, K. D., & Fakhri, S. A. K. (2003). Introduction: Transnationalism and its edges. *Ethnic and Racial studies*, 26(2), 207–217.
- Yonezawa, Y., & Yonezawa, A. (2016). Internationalization of higher education as a response to globalization: Japan's policy challenges since the 1980s. In E. Healy, D. Arunachalam, & T. Mizukami (Eds.), *Creating Social Cohesion in an Interdependent World* (pp. 191–204). New York: Palgrave Macmillan.
- Young, H. R., Glerum, D. R., Wang, W., & Joseph, D. L. (2018). Who are the most engaged at work? A meta-analysis of personality and employee engagement. *Journal of Organizational Behavior*, 16, 1–17.
- Yu, W. (2017). Chinese student migrants in transition: A pathway from international students to skilled migrants. In Van Riemsdijk, & Q. Wang (Eds.), *Rethinking international skilled migration* (pp. 93–110). Routledge.
- Van Der Zee, K. I., & Van Oudenhoven, J. P. (2000). The Multicultural Personality Questionnaire: A multidimensional instrument of multicultural effectiveness. *European Journal of Personality*, 14(4), 291–309.
- Zhang, J., & Goodson, P. (2011). Predictors of international students' psychosocial adjustment to life in the United States: A systematic review. *International Journal of Intercultural Relations*, 35(2), 139–162.
- Zheng, Z., & Xie, Y. (2004). Asian Americans' earnings disadvantage reexamined: The role of place in education. *American Journal of Sociology*, 109, 1057–1108.
- Zschirnt, E., & Ruedin, D. (2016). Ethnic discrimination in hiring decisions: a meta-analysis of correspondence tests 1990–2015. *Journal of Ethnic and Migration Studies*, 42(7), 1115–1134.